

BACHARELADO EM
ENGENHARIA DE SOFTWARE

**ACCELERATION OF PATTERN MATCHING
ALGORITHMS USING PARALLEL PROGRAMMING**

MATHEUS ANTÔNIO DE CASTRO DE BARROS

Brasília - DF, 2025

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Qualification Exam document submitted as a partial requirement for the Bachelor's Degree in Software Engineering at the Instituto Brasileiro de Ensino, Desenvolvimento e Pesquisa (IDP).

Orientador

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Aprovado em 08/05/2025

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ABSTRACT

The Aho-Corasick algorithm is a cornerstone of signature-based network intrusion detection systems and malware analysis tools, because of its linear-time complexity in the size of the input. Its state-transition dependency, however, ties the canonical implementation to sequential execution, leaving modern multi-core processors underused. This work designs, implements, and evaluates data-parallel scanning strategies built on POSIX Threads for shared-memory multi-core CPUs. The input text is partitioned into overlapping segments that worker threads scan concurrently over a shared, strictly read-only automaton, keeping the hot path free of locks; a flattened, contiguous output layout is also proposed to eliminate the pointer-chasing cost of emitting matches. The study deliberately targets the memory-bound regime where the automaton greatly exceeds the last-level cache, using realistic Snort and Emerging Threats rule sets (up to roughly 507 MiB of automaton) over multi-gigabyte corpora on a hybrid Intel Core i5-1235U processor. The results show that text-level data parallelism outperforms pattern-set partitioning: search speedups reach up to $4.79\times$ on twelve threads for a moderate dictionary and $4.52\times$ at the multi-gigabyte scale, while the flattened layout adds a regime-dependent gain that grows with memory pressure, up to +13.0% on a single thread. When the automaton is forty-two times larger than the cache, however, throughput drops by 74.0% and the parallel speedup peaks near $2.85\times$ at six threads on the canonical corpus before regressing, reaching $2.79\times$ at twelve threads on the multi-gigabyte corpus, exposing memory bandwidth as the physical ceiling. End-to-end accelerations reach up to $4.48\times$. The central finding is that the dominant barrier to scaling Aho-Corasick for intrusion detection is microarchitectural, namely cache overflow and DRAM bandwidth, rather than the matching logic itself.

Keywords: Aho–Corasick, Parallel Programming, Parallel Computing, Multi-pattern Matching, YARA, Malware Detection, POSIX Threads, Intrusion Detection Systems .

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1

1

INTRODUCTION

Advancements in semiconductor technology have historically driven the growth of computational power. For decades, the number of transistors on a chip doubled roughly every eighteen months, an observation known as *Moore's Law* [1]; rising clock speeds translated these denser circuits into proportional single-core performance gains. However, in the mid-2000s, the hardware industry encountered significant physical barriers, such as excessive power consumption and challenges in heat dissipation, which limited the increase in processor clock frequency.

The response to these technological limitations was a change in processor design: instead of focusing on making a single core faster, the emphasis shifted to integrating multiple processing cores onto a single chip. Thus, multi-core processors emerged as the dominant architecture in modern computers. This transition opened up new possibilities for performance enhancement; however, to take advantage of multi-core processors, programs must be intentionally written or adapted to execute tasks in parallel [1].

This architectural shift from sequential to parallel execution has a profound impact on high-performance applications. In many scenarios, such as bioinformatics pipelines and network intrusion detection, processing large data volumes efficiently is essential. These applications rely on highly efficient multi-pattern-matching algorithms, and the Aho-Corasick algorithm stands out for its efficiency. Widely adopted in security tools — including YARA for malware classification and Snort and Suricata for signature-based network intrusion detection [2] — the Aho-Corasick algorithm serves as a backbone for multi-pattern search engines [3].

Despite its efficiency, the Aho-Corasick algorithm is intrinsically sequential: its automaton processes input strictly left-to-right, with each state transition depending on the preceding one. As a result, when scanning a single large input on a modern multi-core CPU, the entire computational load is concentrated on one core. This serial dependency becomes a major performance bottleneck when scanning large target data, such as full disk images, packet captures, or genomic sequences. Consequently, the disparity between the algorithm's sequential nature and modern hardware's parallel processing capabilities widens as data volumes grow [4].

1.1 MOTIVATION

The intrinsically sequential nature of Aho-Corasick, combined with the prevalence of multi-core architectures in general-purpose computing, presents a clear opportunity for acceleration through parallel computing. Previous studies report speedups on specialized platforms and in specific application domains [2, 5]; however, the behavior of a shared-memory, multi-threaded implementation on commodity CPUs in the *memory-bound regime* — where the automaton size far exceeds the last-level cache — has received comparatively little attention. This is particularly true for modern hybrid processors that combine high-performance and high-efficiency cores.

As pattern sets for tools such as Snort and Suricata grow in size, the corresponding Aho-Corasick automata can occupy hundreds of megabytes, pushing the hot data structure well beyond the cache hierarchy [6]. In this regime, the primary performance bottleneck shifts from computation to memory bandwidth. The interaction of multiple threads sharing the same last-level cache changes the scaling behavior in ways that compute-bound models do not capture. Understanding these limits on realistic workloads is therefore a prerequisite for deploying parallelized pattern-matching engines in production NIDS pipelines.

This thesis addresses that gap through the following research questions:

- **RQ1:** What throughput and speedup gains does text-level data parallelism, implemented with POSIX Threads over a shared read-only automaton, yield for Aho-Corasick on a shared-memory multi-core CPU across different workload and pattern-set configurations?
- **RQ2:** How does the automaton-to-last-level-cache size ratio constrain parallel scalability, and what is the dominant performance ceiling in the memory-bound regime on a hybrid P/E-core processor?

1.2 OBJECTIVES

The general objective of this thesis is to design, implement, and evaluate a parallel version of the Aho-Corasick algorithm targeting shared-memory multi-core CPU architectures, using the POSIX Threads programming model, in order to accelerate multi-pattern matching on large inputs.

To achieve this general objective, the following specific objectives are defined:

1. To design and implement a parallel version of the Aho-Corasick algorithm in C, using the pthreads API, exploiting intra-input parallelism on shared-memory multi-core CPUs;
2. To conduct a rigorous experimental evaluation that quantifies the performance

gains of the parallel implementation in terms of speedup, throughput, and parallel efficiency, against a sequential baseline;

3. To analyze the scalability of the proposed solution under different workload, pattern-set, and thread-count configurations.

1.3 STRUCTURE OF THE THESIS

This document is organized as follows. Chapter 2 presents the Theoretical Framework, introducing the fundamentals of parallel computing, including architectures, performance metrics, and thread-based programming models, with emphasis on the POSIX Threads API. It also discusses the core data structures and algorithms relevant to this work, including Tries, the general pattern-matching problem, and a detailed analysis of the Aho-Corasick algorithm. Chapter 3 describes the Systematic Literature Review, presenting the research questions, search strategy, and selection criteria adopted to identify the state of the art and define the research gap. Chapter 4 presents the Proposed Solution, detailing the parallel-scanning strategy, the boundary-overlap and load-balancing mechanisms, and the flattened output layout that constitute the design. Chapter 5 describes the Methodology, specifying the experimental environment, datasets, metrics, measurement protocol, correctness validation, and the hypotheses that the experiments are designed to test. Chapter 6 reports and discusses the Results, isolating the contribution of each optimization and positioning the findings against the related literature. Finally, Chapter 7 presents the Conclusion, summarizing the contributions, the threats to validity, and directions for future work.

2

2

THEORETICAL
WORK

FRAME-

2.1 FUNDAMENTALS OF PARALLELISM

Parallelism in computing refers to the simultaneous execution of multiple tasks or parts of the same task, with the primary goal of reducing a program's total execution time. It is crucial to distinguish parallelism from concurrency. Concurrency occurs when multiple tasks are in progress at a given time, but may not be executing simultaneously. A single-core system can exhibit concurrency by rapidly switching between the execution of different tasks, simulating parallelism through preemption. True parallelism, on the other hand, requires hardware with multiple processing units [7].

2.1.1 Parallel Computing Architectures

Parallel computer architectures are classified according to various criteria. A prominent taxonomy, proposed by Flynn [8], categorizes these architectures based on instruction and data streams:

- **SISD (Single Instruction, Single Data):** Represents a conventional single-processor architecture, where a single instruction operates on a single data stream;
- **SIMD (Single Instruction, Multiple Data):** Involves multiple processing units that execute the same instruction concurrently on different data elements;
- **MISD (Multiple Instruction, Single Data):** Characterizes systems where multiple processing units apply distinct instructions on the same data stream;
- **MIMD (Multiple Instruction, Multiple Data):** Involves multiple processing units that execute distinct instructions on distinct data streams independently. This paradigm is directly related to multi-core architectures.

Within the MIMD category, systems are further distinguished based on their memory organization [1]:

- **Shared-Memory Systems:** In these systems, multiple processors share a physical memory address space. Inter-processor communication (IPC) is typically achieved by reading from and writing to shared variables in memory or through

message passing mechanisms. Multi-core processors are the most common and widely available example of this architecture, where all cores on a chip share access to main memory;

- **Distributed-Memory Systems:** Each processor in these systems has its own locally accessible memory, and communication between processors is usually done by exchanging messages across an interconnection network. Computer clusters are examples of this architectural model.

The present work focuses on the acceleration of algorithms in environments with multi-core processors, which fall into the shared memory MIMD category. A multi-core processor (MCP) is a single chip that contains two or more central processing units (CPUs) called 'cores'. Each of these cores is capable of reading and executing program instructions independently, essentially functioning as a complete processor.

2.1.2 Processes vs Threads

Within shared-memory systems, such as multi-core processors, software-level parallelism is commonly achieved through the execution of multiple threads within a single process.

A process is an instance of an executing program, an entity possessing its own dedicated virtual address space, file descriptors, environment variables, and other Operating System (OS) resources, all isolated from other processes [1].

A thread, on the other hand, is a lighter unit of execution started by a process. The same process can control, concurrently or in parallel, multiple threads. A key characteristic is that all threads within a process share its virtual address space, which encompasses executable code, global data, and the heap segment [1]. However, each thread maintains its private resources, which are essential for its independent execution context:

- A unique identifier (thread ID);
- A set of CPU registers, including the program counter (PC);
- A separate execution stack, used for local variables and function call control;
- Thread-specific state, such as signal masking and scheduling priority.

2.1.3 POSIX Thread Model (Pthreads)

The POSIX Thread Model, commonly referred to as "Pthreads", is a standard that defines an application programming interface (API) for thread creation and management, as specified by IEEE Std 1003.1c-1995. Pthreads provides a portable suite of data types, functions, and constants that allows developers to create multithreaded

applications that are compilable and executable across diverse POSIX-compliant operating systems (e.g., Linux, macOS, and various Unix variants). The Pthreads API provides functionality for managing the thread lifecycle—including creation, termination, and joining—and, critically, incorporates synchronization primitives. These primitives, such as mutexes and condition variables, are essential for managing access to shared resources within shared-memory environments [7] [1].

2.1.4 Cache Architecture and Memory-Bound Parallelism

Modern shared-memory processors organize their caches in a hierarchy. Each core has private L1 and L2 caches—typically tens to hundreds of kilobytes—for low-latency access to recently used data. A shared last-level cache (LLC), often several megabytes, is accessible to all cores on the chip and serves as the final caching tier before main memory [1].

A workload is *compute-bound* when its execution time is dominated by arithmetic operations, leaving memory bandwidth underutilized. It is *memory-bandwidth-bound* (or simply *memory-bound*) when the bottleneck is the rate at which data can be fetched from DRAM rather than computational throughput. Pattern-matching algorithms that operate on large automata—whose transition tables exceed the LLC capacity—fall into this regime: each state transition triggers a cache miss that must be resolved from DRAM, making throughput a function of available memory bandwidth rather than CPU frequency or core count [9].

Parallelism can intensify memory pressure. When multiple threads simultaneously traverse different regions of a large, read-only automaton, the combined working set exceeds the LLC, raising the aggregate cache-miss rate for all threads. Two additional hazards arise when threads share writable data. *Cache coherence* protocols (such as MESI) ensure that modifications made by one core become visible to all others; they introduce inter-core invalidation traffic whenever a shared cache line is written [1]. *False sharing* occurs when two threads write to logically independent variables that happen to reside on the same cache line, causing coherence invalidations despite the absence of a true data dependency. Aligning per-thread accumulators to cache-line boundaries eliminates false sharing.

Non-Uniform Memory Access (NUMA) architectures differentiate memory latency based on which socket owns the physical memory being accessed. Although the experimental platform used in this work—the Intel i5-1235U—employs a single-socket Uniform Memory Access (UMA) topology, NUMA is an important consideration in multi-socket server deployments where memory-bound workloads exhibit significant latency asymmetry [9].

2.1.5 Hybrid P/E-Core Architectures

Modern consumer and mobile processors have introduced a *hybrid* microarchitec-

ture that integrates two distinct core types on a single die. Intel's Alder Lake platform—which includes the Intel Core i5-1235U used as the experimental platform in this work—pairs two high-performance cores (P-cores), each supporting Hyper-Threading (two logical hardware threads per core), with eight energy-efficient cores (E-cores), each providing a single hardware thread, for a total of twelve logical hardware threads.

P-cores feature wider out-of-order execution pipelines and higher operating frequencies, delivering greater per-thread throughput at higher power consumption. E-cores are physically smaller and optimized for throughput-per-watt rather than single-thread latency; they operate at lower frequencies. A memory-bound workload therefore achieves significantly higher per-thread throughput on a P-core than on an E-core.

This heterogeneity shapes parallel scaling curves in a characteristic way. At low thread counts, threads are preferentially dispatched to P-core hardware threads; speedup rises steeply because each additional thread contributes high-throughput capacity. Once all P-core hardware threads are occupied—four on the i5-1235U—additional threads are scheduled onto E-cores, which contribute lower per-thread throughput. The cumulative scaling curve therefore exhibits a *knee* near the P-core saturation point, beyond which the marginal speedup per additional thread declines. This architectural asymmetry underpins the scaling patterns discussed in the Results and Discussion chapter.

2.2 PARALLEL PERFORMANCE EVALUATION METRICS

Performance evaluation is fundamental to understanding the effectiveness of a parallel program. The following metrics compare the execution time of the parallel program with its equivalent sequential counterpart [9].

2.2.1 Execution Time and Overhead

The performance of a program is measured by its execution time. For parallel programs, the following terms are defined:

- **T_s**: The execution time of a baseline sequential algorithm executed on a single processor unit.
- **T_p**: The execution time of the parallel program utilizing p processors.

Ideally, T_p is expected to be substantially less than T_s . However, parallel execution introduces *overhead*. This encompasses the time spent by processors on tasks that do not directly contribute to problem resolution, including communication, synchronization, load balance, and other computations inherent to the parallel implementation [9].

2.2.2 Speedup

Speedup (S) measures the performance gain achieved through parallel execution relative to sequential execution, as shown in Eq. 1. It is defined as the ratio of sequential

execution time to parallel execution time:

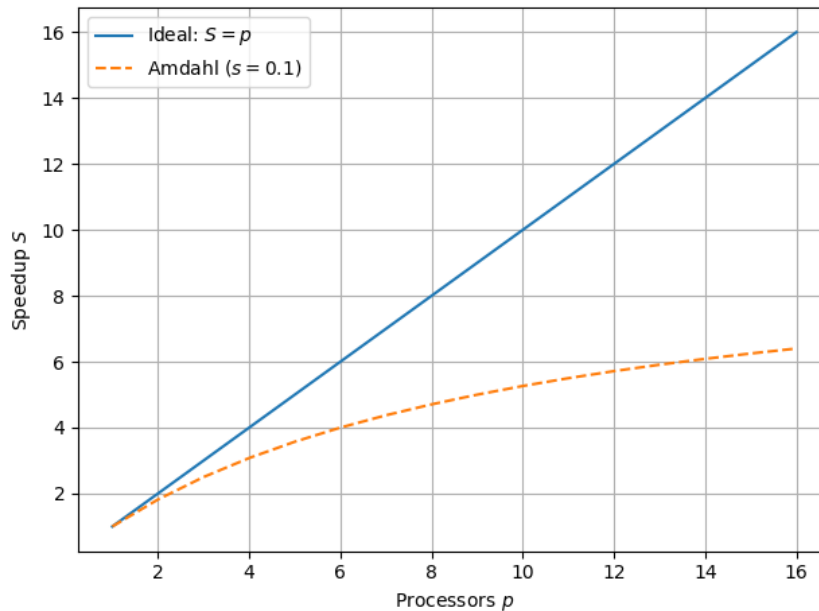
$$S = \frac{T_s}{T_p} \quad (1)$$

An ideal, or linear, speedup would be $S = p$. Amdahl's Law establishes a theoretical upper bound on the speedup achievable by parallelizing a program with a fixed problem size. If s represents the fraction of a program's sequential execution time that is inherently serial (non-parallelizable), and f is the parallelizable fraction ($s + f = 1$), the maximum speedup on p processors is given by Eq. 2.

$$S \leq \frac{1}{s + f/p} \quad (2)$$

In the limit, as the number of processors (p) tends to infinity, the maximum speedup is bounded by $\frac{1}{s}$. This implies that even a small sequential fraction (small s) can severely limit the total speedup, regardless of the number of processors available. Figure 1 illustrates the theoretical speedup according to Amdahl's Law compared to ideal speedup for a given sequential fraction.

Figure 1: Ideal vs. Amdahl's theoretical speedup ($s = 0.1$).



A complementary perspective is offered by the Gustafson-Barsis Law in Eq. 3. This law argues that an approximately linear speedup is achievable for problems that benefit from this scalability. This formulation suggests that for problems that can be scaled, it is possible to obtain an approximately linear speedup with the number of processors, even in the presence of a serial fraction. Both Amdahl's and Gustafson's laws are important for understanding the limits and potential of parallelism in different scenarios [1].

$$S_{\text{scaled}} = s + (1 - s) \cdot p \quad (3)$$

2.2.3 Efficiency

Efficiency (E) measures the effectiveness of processor utilization in a parallel program. It is defined as the ratio of speedup to the number of processors p , as in Eq. 4:

$$E = \frac{S}{p} = \frac{T_s}{p \cdot T_p} \quad (4)$$

Efficiency values range between 0 and 1 (inclusive), often expressed as a percentage (0% to 100%). An efficiency of 1 (or 100%) signifies ideal, linear speedup, implying that all processors are contributing productively throughout the execution. In practice, efficiency typically diminishes as the processor count increases, primarily due to the escalating impact of parallel overhead [9].

2.2.4 Scalability

In parallel computing, scalability denotes a system's ability to sustain high performance (often via efficiency) as both the number of processors and/or the problem size grow [9].

- **Strong Scalability:** Measures how execution time changes when the processor count p increases for a fixed total problem size, denoted W (for example, the total number of array elements to process). The goal is to keep efficiency nearly constant as p increases. Amdahl's Law directly applies in this scenario;
- **Weak Scalability:** Measures how execution time changes when both the processor count p and the total problem size W increase such that the work per processor (W/p) stays constant. The aim is to maintain efficiency as p and W grow proportionally. Gustafson's Law is most relevant here.

A parallel system is considered scalable if its efficiency remains above an acceptable threshold as both the number of processors p and the problem size W increase.

2.3 TRIES (PREFIX TREES)

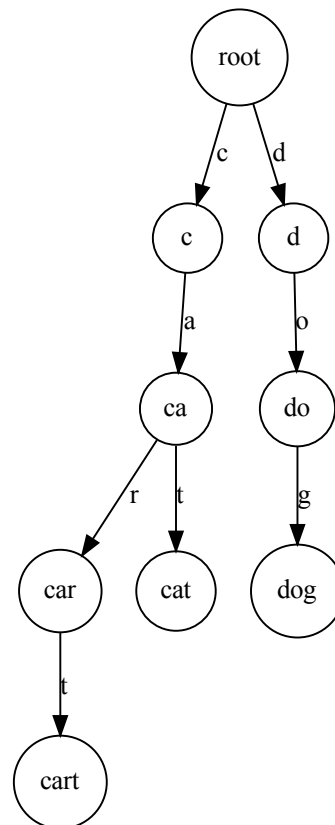
This section explains the Trie data structure, detailing its defining properties, core operational mechanics, performance aspects, and its significance as a foundational element in advanced string processing algorithms, particularly the Aho-Corasick algorithm.

2.3.1 Definition and Basic Structure

A Trie—also known as a prefix tree—is a specialized tree-like data structure optimized for the efficient storage and retrieval of a dynamic set of strings. Edward Fredkin

introduced the term “trie” in 1960, highlighting its primary application in information retrieval [10], [11]. Figure 2 illustrates an example of a trie storing a set of words.

Figure 2: Example trie storing {“car”, “cat”, “cart”, “dog”}. Nodes marking the end of a complete word are shown with a double circle.



Source: Prepared by the author.

Tries store strings by organizing nodes so that each path from the root to a node represents a unique prefix. In a standard R -way trie, each node can branch to R children, where R denotes the size of the character alphabet. The existence of a parent-child edge highlights two key aspects of tries:

- The edge’s position (index) within the parent node’s array of R possible child links directly corresponds to a unique character in the alphabet. For instance, index 0 might correspond to ‘a’, index 1 to ‘b’, and so on, according to a predefined mapping.
- The character associated with this edge extends the prefix represented by the parent; this implies that at least one word (or key) in the trie contains this character sequence.

To signify that such a prefix also constitutes a complete key present in the trie, the node reached at the end of this path is typically marked by storing an associated non-null value or a boolean flag. The characters themselves are implicitly encoded by the trie's structure rather than being explicitly stored within each node [12].

2.3.2 Core Operations

Tries support several primary operations, including search, insertion, and deletion [12]. These operations are explained below:

- **Search:** locating a key involves traversing the trie starting from the root, following the successive characters of the target key. Each character dictates which child link to follow. The search operation succeeds if a complete path corresponding to all characters of the key exists and the node reached at the end of this path is appropriately marked as representing a complete key. The search fails if, at any point, a required link is null (i.e., no path for the current character exists) or if the path is fully traversed but the final node is not marked as a complete key;
- **Insertion:** the insertion process begins similarly to a search, by traversing the trie according to the characters of the key to be added. If the path for the key does not extend to its full length (i.e., a null link is encountered), new nodes are created and linked to form the remainder of the path. The node corresponding to the final character of the inserted key is then marked to indicate that it is a complete key in the set;
- **Deletion:** deleting a key first involves locating the node corresponding to the key's final character and removing its end-of-word marker. If there are no child links (meaning it is not a prefix for any other key), it can be deleted. This deletion may proceed recursively: when a child node is removed, if its parent has no other children and the parent node is not marking a key, the parent node becomes redundant and can be deleted. This process continues up to the root as long as parent nodes become redundant.

2.3.3 Advantages and Performance Characteristics

Tries provide distinct advantages for operations on string collections, as detailed below.

- **Time Complexity:** a primary advantage is that the time required for search and insertion operations depends primarily on the length of the key (L), not on the total number (N) of keys stored. Specifically, these operations typically require examining a number of nodes proportional to L . [12] state that the number of array accesses (node visits) is at most $L + 1$ [12]. This $O(L)$ performance makes tries

highly efficient for prefix-based searches (e.g., retrieving all keys starting with a given prefix) and ensures that search performance does not substantially degrade as the dataset size increases;

- **Prefix Sharing:** when stored strings share common prefixes, tries inherently exploit this by having those strings share the initial path from the root. This sharing can lead to significant space savings compared to storing each string independently, particularly if there is substantial prefix overlap among keys [10].

While prefix sharing can save space, the overall space complexity of R -way tries requires careful consideration. Each node in an R -way trie conceptually provides R potential links. If the alphabet size R is large (e.g., for Unicode) and the actual branching factor at most nodes is low (a sparse trie), allocating space for all R links per node can lead to considerable memory consumption due to numerous unused (null) links [11], [12].

2.3.4 Foundational Role in String Processing

The trie data structure is a cornerstone for numerous advanced string processing algorithms. Critically for this work, the Aho-Corasick algorithm, engineered for the efficient, simultaneous matching of multiple patterns within a text, employs a trie—constructed from the set of patterns (the dictionary)—as its initial structural framework [4]. A thorough grasp of trie principles is thus essential for developing and analyzing the Aho-Corasick algorithm.

2.4 PATTERN MATCHING

This section introduces the fundamental problem of pattern matching within strings, explores key algorithmic approaches to its solution, and highlights its significance across various computational domains. This foundational discussion provides the essential context for understanding advanced multi-pattern matching algorithms, such as the Aho-Corasick algorithm. In general information processing, pattern (or *string*) matching underpins the functionality of text editors and web search engines for keyword searches. The field of computational biology extensively utilizes pattern matching for analyzing genetic sequences [12]. Furthermore, in the field of information security, pattern matching is crucial: Network Intrusion Detection Systems (NIDS) utilize it to identify malicious signatures in network traffic. Tools like YARA, for instance, define rules based on textual or binary patterns to classify and identify malware [3].

2.4.1 The Pattern Matching Problem

The *string matching* problem consists of determining whether a string P (the *pattern*) of length M occurs within a string T (the *text*) of length N , where $M \leq N$. Formally, it seeks all positions pos such that the condition in Eq. 5 holds for every $i \in \{0, 1, \dots, M -$

1}, and each valid pos corresponds to an occurrence of the pattern in the text:

$$P[i] = T[pos + i], \quad \text{with } pos + i < N \quad (5)$$

Beyond simply finding the first match, the problem often extends to locating all occurrences, enumerating them, or extracting surrounding contextual information for each match [12].

Algorithms that address the string matching problem are generally categorized based on the number of patterns they process and the exactness of the match. This thesis focuses on *Exact Matching*, where the pattern must correspond precisely to a segment of the text. A key distinction for this study lies between algorithms designed for *Single-Pattern Matching* and those optimized for *Multi-Pattern Matching*, where the objective is to identify all occurrences of any pattern from a predetermined set of patterns within a given text [4], [13].

2.4.2 The Naive Algorithm

The naive or brute-force algorithm represents the most straightforward method for exact pattern matching. Its operation involves iteratively aligning the pattern P with the text T at every possible starting position i (from 0 to $N - M$). For each alignment, P is compared character by character, from left to right, against the corresponding substring. If all characters match, an occurrence is reported [13].

Following this comparison (whether a match or a mismatch), the algorithm shifts P one position to the right relative to T and proceeds to the next alignment. Despite its simplicity, Knuth, Morris, and Pratt (1977) state that this method can be highly inefficient [14]. Its worst-case time complexity is $O(M \cdot N)$ character comparisons. This occurs, for example, when searching for a pattern like $P = a^nb$ within a text $T = a^{2n}b$, where many partial matches occur. To address these limitations, more advanced pattern matching algorithms include a preprocessing phase for the pattern P . This phase extracts structural information about P that is then utilized during the search, enabling larger shifts of the pattern relative to the text and thus reducing the total number of character comparisons.

2.4.3 Knuth-Morris-Pratt (KMP) Algorithm

The KMP algorithm [14] efficiently resolves the pattern matching problem with an optimal worst-case time complexity of $O(N + M)$. Unlike the naïve approach, KMP avoids re-checking characters that have already been matched. To achieve this, KMP first preprocesses the pattern P by constructing a special array known as the *failure array* (or *prefix array*), typically denoted by $p[1..M]$. This array indicates the length of the longest prefix that is also a suffix for each prefix of P . With this information, when a

mismatch occurs during the search, the algorithm utilizes the failure array to determine how far the pattern should shift to the right, bypassing unnecessary comparisons and preventing redundant checks.

2.4.4 Boyer-Moore (BM) Algorithm

The Boyer-Moore (BM) algorithm [15] is widely recognized for its excellent practical performance, often outperforming linear-time methods in typical cases. Unlike the KMP algorithm, Boyer-Moore typically compares characters starting from the end of the pattern and moves backward. Its efficiency primarily stems from two rules (or heuristics):

- **Bad Character Rule:** When a mismatch occurs, this rule uses the character from the text that caused the mismatch to determine how far the pattern can safely shift. The algorithm shifts the pattern to align this mismatched text character with its last occurrence in the pattern or moves completely past the mismatched character if it does not appear in the pattern;
- **Good Suffix Rule:** When a mismatch happens after matching one or more characters from the end of the pattern, this rule uses the matched suffix to decide how far to shift the pattern. It searches for another occurrence of the matched suffix within the pattern, ensuring that no previously matched characters are unnecessarily compared again.

These two rules combined enable the Boyer-Moore algorithm to skip larger portions of the text, significantly reducing the total number of comparisons.

2.4.5 The Need for Specialized Multi-Pattern Algorithms

When the task involves searching for multiple patterns simultaneously from a set $K = \{P_1, P_2, \dots, P_k\}$ within a text T , simply applying a single-pattern matching algorithm for each P_i independently proves highly inefficient. This “iterative” approach requires scanning the text multiple times, resulting in a total time complexity that can be approximately $O(k \cdot (N + M_{\text{avg}}))$ or worse, where M_{avg} is the average pattern length [4].

To efficiently handle applications that require simultaneous searches for a large number of patterns, specialized algorithms are essential. These methods usually preprocess the entire set of patterns to create a unified data structure, which enables a single, efficient pass over the text to identify all occurrences of any pattern within the set [13].

The theoretical framework of *finite automata* provides a robust and elegant solution to this problem. By constructing a single *finite automaton* (FA) that can recognize all patterns in a given set, the text can be processed in a single continuous scan. Each character read from the text triggers a *state transition* within the FA. When the FA reaches an *accepting state*—a state explicitly designed to indicate the successful recognition of

one or more patterns—the algorithm reports an occurrence. Algorithms like the Aho-Corasick algorithm exemplify this approach; they automate the construction of such a *pattern-matching machine*, typically based on a trie structure augmented with specialized *failure transitions* [4]. This automaton-based paradigm forms the critical foundation for efficiently solving multi-pattern matching problems.

2.5 THE AHO-CORASICK ALGORITHM

This section provides a comprehensive examination of the Aho-Corasick algorithm, a highly efficient string-matching method designed to locate all occurrences of multiple patterns (keywords) within a given text. It combines *automata theory* with *trie* data structures to efficiently solve the *multi-pattern search* problem. Its search phase runs in $O(n + z)$ time, where n is the text length and z is the number of matches [4].

2.5.1 Context

The task of concurrently identifying all instances of a large collection of patterns within a text poses a significant computational challenge. A naïve approach—running a *single-pattern matching* algorithm such as KMP or BM for each of the k patterns—requires repeated scans of the text and has a worst-case time complexity of $O(k \cdot n)$, where k is the number of patterns and n is the text length, rendering it impractical for large pattern sets or long texts [13].

The Aho-Corasick algorithm overcomes this by building one *finite automaton* that encodes all patterns simultaneously. This automaton processes the input text in a single pass to detect all matches. The algorithm effectively combines the *state-transition* efficiency of finite automata with the prefix-sharing advantages of tries, following *failure links* on mismatches and resuming matching without rescanning characters—an extension of KMP’s *failure-function* idea to a trie of patterns [13].

2.5.2 Algorithm Components and Construction

Aho-Corasick has two main phases: (i) construction, which builds the automaton, and (ii) search, which scans the text. The automaton, introduced in [4], has three key tables, built in the following order:

- **Keyword Trie (goto):** a *keyword trie* is first constructed from the pattern set $K = \{P_1, P_2, \dots, P_k\}$. Each node represents a state, and each edge corresponds to a character transition. A path from the root to any state s spells a prefix of one or more keywords. Formally, the *goto function* $\text{goto}(s, a)$ returns the next state s' when an edge labelled a exists; otherwise the transition fails. Undefined transitions from the root are assigned back to the root, i.e., $\text{goto}(0, a) = 0$, creating implicit self-loops for every alphabet symbol;
- **Failure Function (f):** for any state s (except the root), $f(s)$ is the state s' whose

path label is the longest proper suffix of the path label of s that is also a prefix of some keyword. On a mismatch, the automaton follows $f(s)$ and resumes matching without consuming the current character. The *failure function* is computed level-by-level (breadth-first) after the goto table is complete;

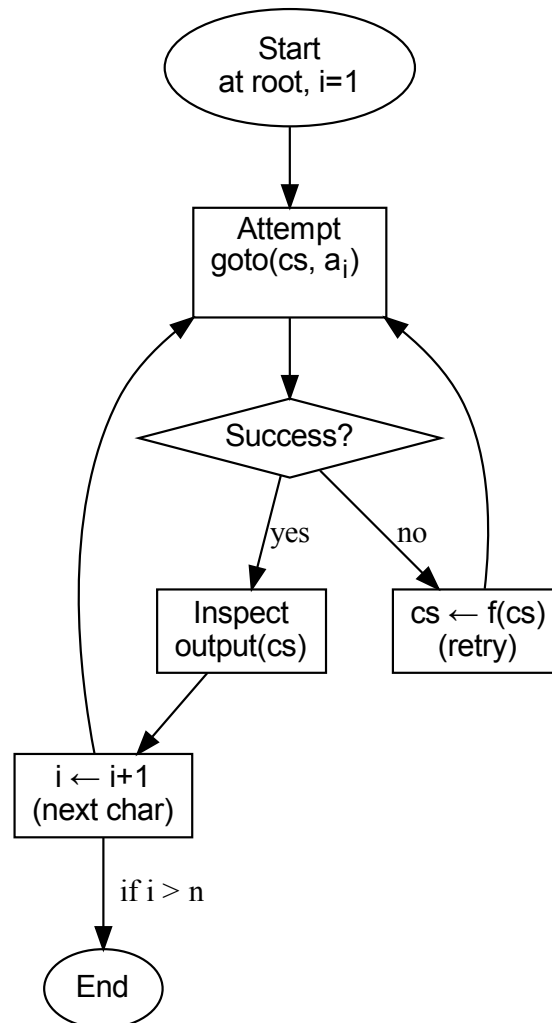
- **The Output Function:** the *output function*, $\text{output}(s)$, identifies all keywords that end at state s . For a given state s , $\text{output}(s)$ is the set of all patterns P_i in K such that the string corresponding to the path to s is P_i . To ensure all matches are reported, including those that are suffixes of other patterns, $\text{output}(s)$ is augmented during the failure function computation: if $f(s) = s'$, then $\text{output}(s) \leftarrow \text{output}(s) \cup \text{output}(f(s))$. This incremental union lets each state report every pattern that ends either at that state or at any suffix state reached through its failure chain.

2.5.3 Matching Phase

Once the Aho–Corasick automaton has been built, it scans the text $T = a_1a_2 \dots a_n$ in one left-to-right pass. At each position i , the machine attempts to follow the transition labelled a_i from its *current_state* (cs); if that edge exists, it moves along it, otherwise it follows *failure links* (f) until either a matching transition is found or the root is reached. Every time a successful transition is made, the automaton inspects the output set of the new state and emits all patterns ending at position i .

Figure 3 illustrates this control flow in detail, which: starts at the root with $i = 1$, attempt each $\text{goto}(\text{current_state}, a_i)$, follow failure links on a “no” decision, report matches via $\text{output}(\text{current_state})$ on a “yes,” advance i , and terminate when $i > n$.

Figure 3: Flowchart of the Aho–Corasick matching phase.



Source: Prepared by the author.

Because each character can trigger at most one eventual successful goto and any number of failure-link traversals, the total number of transitions (goto plus failure) over the entire scan is bounded by $< 2n$ [4].

2.5.4 Time and Space Complexity

The Aho-Corasick algorithm exhibits optimal *linear time complexity* for multi-pattern matching [4, 13]. The complexities for the construction and search phases and storing the automaton are as follows:

- **Construction phase:** building the goto, failure, and output functions takes $O(L)$ time, where L is the total length of all keywords. If a dense array is used for the root, an additional $O(|\Sigma|)$ initialization cost is incurred, giving $O(L + |\Sigma|)$;

- **Search phase:** Scanning a text of length n performs fewer than $2n$ state transitions. Reporting matches costs time proportional to the total number of occurrences z , so the total search time is $O(n + z)$;
- **Space complexity:** Storing the automaton requires $O(L + |\Sigma|)$ space: $O(L)$ for trie edges, $O(L)$ pointers for failure links, and additional space for output lists (proportional to the number of stored outputs).

2.5.5 NFA versus DFA Representations

The automaton described above is, strictly speaking, a *nondeterministic finite automaton* (NFA): the goto function is undefined on many (state, character) pairs, and the search algorithm resolves those gaps by traversing failure links at runtime, requiring multiple memory accesses per character in the worst case. A widely adopted optimization converts this NFA into a *deterministic finite automaton* (DFA) by precomputing the failure-link chains during construction: for every state s and character a , the full deterministic transition $\delta(s, a)$ is stored in a flat two-dimensional table of size $|Q| \times |\Sigma|$, where $|Q|$ is the number of states and $|\Sigma|$ is the alphabet size [4, 13]. With this representation every character in the input triggers exactly one array lookup; no runtime failure-link traversal occurs, which reduces per-character latency and simplifies the inner search loop.

The cost of the DFA representation is memory: for the standard 256-symbol byte alphabet the flat transition table occupies $4 \cdot |Q| \cdot 256$ bytes. This footprint grows linearly with the pattern-set size and can easily exceed the processor's last-level cache (LLC). When the table no longer fits in cache, each state transition becomes a potential cache miss, shifting the workload from compute-bound to *memory-bandwidth-bound*. The implementation in this work uses the DFA (flat transition table) representation exclusively; the goto function, failure function, and output function described in the preceding subsection are intermediate construction artefacts that are not retained at search time.

2.5.6 Applications and Relevance

The Aho-Corasick algorithm is widely used in applications that require high-performance *multi-pattern matching*, where its single-pass processing of the input against the entire pattern set is essential. Representative use cases include large-scale text analysis, motif searching in DNA or protein sequences [5], spam filtering, Network Intrusion Detection Systems (NIDS) that scan traffic against thousands of signatures, and anti-malware tooling—YARA, for example, employs Aho-Corasick to match the literal string components extracted from its rules [3].

While Aho-Corasick excels at exact literal matching, systems like NIDS often layer additional capabilities—such as *regular-expression matching*—on top of Aho-Corasick

engines to support more expressive signatures, which affects overall performance characteristics. The algorithm therefore remains a theoretically elegant and practically impactful example of *finite-automata theory* applied to string processing [13], [3].



3

3

SYSTEMATIC REVIEW

This section outlines the methodology employed for the Systematic Literature Review (SLR) conducted to identify, evaluate, and synthesize pertinent studies concerning the parallelization and optimization of the Aho-Corasick algorithm. The review focuses on implementations targeting multi-core CPUs. The objective of this SLR is to support the development of the present research and to answer the formulated research questions.

3.1 RESEARCH QUESTIONS

The following Research Questions (RQs) were formulated to guide the literature search, selection, and analysis process:

1. **RQ1:** What parallelization techniques and associated challenges are documented in the literature for implementing the Aho-Corasick algorithm utilizing threads on multi-core CPUs?
2. **RQ2:** What specific optimizations for the Aho-Corasick algorithm are employed in application scenarios involving extensive pattern sets, particularly within domains of information security?

3.2 SEARCH STRATEGY AND STUDY SELECTION PROTOCOL

The search and selection protocol for identifying primary studies encompassed the definition of data sources, construction of search queries, and establishment of inclusion and exclusion criteria.

3.2.1 Data Sources

The following scientific databases were selected due to their scope and relevance in the area of Computer Science and Software Engineering:

- Web of Science (WoS)
- Scopus
- ACM Digital Library (ACM DL)

- IEEE Xplore

3.2.2 Search Queries

Two primary search strings were constructed to address the aspects of parallelization (RQ1) and optimization/application (RQ2) pertinent to the Aho-Corasick algorithm. The literature search was executed in May 2025.

String 1 (Focus on Parallelization):

"Aho-Corasick" AND (parallel* OR multithread* OR multi-thread* OR thread* OR concurren* OR hpc OR high-perform* OR high perform*) AND ("multi-core" OR multicore OR "many-core" OR CPU) NOT "distributed systems"

String 2 (Focus on Optimization and Applications):

"Aho-Corasick" AND (optimi* OR accelerat* OR perform* OR evaluat* OR improve*) AND ("application" OR "implementation" OR "use case")

3.2.3 Inclusion (IC) and Exclusion (EC) Criteria

The subsequent criteria were established for the selection of primary studies:

Inclusion Criteria (IC):

- **IC1 (Primary Focus):** The study addresses the Aho-Corasick (AC) algorithm or presents parallelization/optimization techniques demonstrably applicable to AC.
- **IC2 (Parallelism Aspect):** The study investigates, proposes, implements, analyzes, or explicitly discusses the parallelization or concurrent execution of the Aho-Corasick algorithm.
- **IC3 (Target Platform):** The study reports empirical results or analyses about multi-core CPU platforms.
- **IC4 (RQ Relevance):** The study provides information relevant to addressing at least one RQ.
- **IC5 (Language):** Published in English or Portuguese.
- **IC6 (Accessibility):** Full-text availability of the study.
- **IC7 (Publication Window):** Published between January 2015 and May 2025, inclusive.

Exclusion Criteria (EC):

- **EC1 (Irrelevant Focus):** Studies with only superficial mentions of the Aho-Corasick algorithm or pattern matching, lacking a primary focus on its parallelization or optimization.

- **EC2 (Non-CPU Platform Exclusivity):** Studies exclusively focused on non-CPU platforms (e.g., GPU, FPGA) without analysis or insights transferable to multi-core CPU environments.
- **EC3 (Insufficient Detail):** Studies lacking sufficient detail regarding methodology, implementation, or reported results.
- **EC4 (Publication Type):** Opinion pieces, editorials, and non-peer-reviewed gray literature.
- **EC5 (Language Barrier):** Publications in languages other than English or Portuguese.
- **EC6 (Duplication):** Duplicate publications (the most comprehensive or recent version was retained).
- **EC7 (Inaccessibility):** Full-text unobtainable.

3.2.4 Selection Process and Initial Search Outcomes

The study selection was a multi-stage process. Initially, the constructed search strings were executed against the selected digital libraries. The retrieved results were subsequently refined using filters provided by each database (e.g., publication period, document type, language, subject area), as detailed in Table 1:

Table 1: Search results and filters applied per database.

Database	String 1 (Found / Kept)	String 2 (Found / Kept)	Filters Applied
ACM Digital Library	122 / 29	274 / 66	2015-2025; Research Article; Publisher: ACM; Title excl.: GPU; Lang: English.
Web of Science	30 / 6	66 / 20	2015-2025; WoS Cats: CS (excl. AI); Title excl.: GPU; Lang: English.
Scopus	34 / 6	145 / 49	2015-2025; Lang: English; Subj: CS, Eng; Types: Article, Conf. Paper; Title excl.: GPU.
IEEE Xplore	25 / 2	60 / 18	2015-2025; Title excl.: GPU; Lang: English.

The Title excl.: GPU filter listed in Table 1 acted only as a coarse identification-stage noise reducer, not as the formal platform criterion: the platform decision is governed by EC2, under which studies on non-CPU platforms (GPU, FPGA, ASIC, SIMD) were retained whenever they offered techniques or findings transferable to shared-memory multi-core CPUs, as argued individually in each review below.

Following the initial database search and filtering stage, the resultant sets of candidate articles were aggregated. The 196 records retained across both search strings

and all four databases (Table 1) were merged, and duplicate entries were removed, yielding 24 unique studies. Each of these was screened by reading its abstract and conclusion against the inclusion and exclusion criteria; the 10 studies most directly pertinent to the research questions were selected as primary studies for the detailed review that follows, while the remaining studies were consulted only as background.

3.3 REVIEW OF SELECTED STUDIES

The selection process described above yielded the corpus of primary studies analyzed in this review. The following subsections examine in detail the ten selected primary studies, spanning shared-memory multi-core parallelizations of Aho–Corasick (RQ1) and optimizations of the algorithm in security-oriented matching workloads (RQ2); each review summarizes the problem addressed, the proposed technique, the experimental evidence, and its relation to the present design.

3.3.1 A Parallel Algorithm of String Matching Based on Message Passing Interface for Multicore Processors

Qu et al. [16] address the limitations of classical multiple-pattern string matching algorithms, such as the Aho–Corasick algorithm, which were originally designed for single-core systems and often become bottlenecks in modern network intrusion detection systems under heavy traffic. The authors highlight the need to use the parallel computation power of modern multicore processors to maintain high performance in deep packet inspection.

To solve this performance bottleneck, the researchers propose a parallel Aho–Corasick string matching algorithm utilizing the Message Passing Interface (MPI) on a shared-memory multicore platform. The core methodology relies on data parallelism: a master thread constructs the automaton sequentially and partitions the input text into contiguous blocks. To prevent missed cross-boundary matches, these blocks overlap by the length of the longest pattern minus one character. The data chunks are then distributed to multiple slave threads via MPI communication, allowing each worker to independently traverse the read-only automaton.

The proposed solution was evaluated on an 8-core multicore system using Snort IDS rulesets of 10,000 and 30,000 patterns. The results demonstrate that the Deterministic Finite Automaton (DFA) variant achieves a peak average throughput of 10.5 Gbps, outperforming the Nondeterministic Finite Automaton (NFA) approach in speed and stability, albeit with higher memory consumption. Furthermore, the evaluation shows that throughput scales monotonically with the addition of CPU cores and thread-level parallelism.

Ultimately, the paper demonstrates that a master-slave data partitioning strategy with overlapping boundaries is highly effective for parallelizing the Aho–Corasick al-

gorithm on multicore processors. However, the reliance on MPI for a single shared-memory node introduces serialized communication overhead, which highlights the advantages of using native shared-memory threading models, such as POSIX Threads, for lock-free coordination.

3.3.2 A Flexible Pattern-Matching Algorithm for Network Intrusion Detection Systems Using Multi-Core Processors

Lee and Yang [2] tackle the throughput bottleneck imposed by Aho-Corasick-based pattern matching in signature-based Network Intrusion Detection Systems (NIDS), where the matching phase can account for up to 70% of total execution time. The authors observe that, although the canonical Aho-Corasick Deterministic Finite Automaton (AC-DFA) offers linear-time scanning, its state-transition table causes heavy memory traffic and frequent last-level cache misses as the pattern set grows. The prior Head-Body Matching (HBM) algorithm mitigates this by partitioning the AC-DFA into a fast head (DFA) and a SIMD-accelerated body (NFA), but its rigid depth-based partitioning criterion typically overshoots or undershoots the cache budget, leading to suboptimal throughput.

To overcome this rigidity, the researchers propose the Flexible Head-Body Matching (FHBM) algorithm. Instead of partitioning the automaton at a fixed depth, FHBM employs a greedy procedure that fills the head with up to a configurable maximum number of states (*H*SIZE), prioritizing parent nodes with the largest number of children and preserving the invariant that all siblings remain on the same side of the partition. The head is represented as a full 256-entry transition table for $O(1)$ lookup, while the body is compacted into an NFA traversed via 128-bit SIMD comparisons. Parallel execution is realized at the thread level: four worker threads, one per physical core, concurrently traverse the read-only Head-Body Finite Automaton (HBFA) on the input stream, exploiting the immutability of the automaton after construction to avoid synchronization in the search phase.

The evaluation was conducted on a quad-core Intel Core i7-3770 (3.4 GHz, 8 MB L3) running 64-bit Ubuntu 14.04, using pattern dictionaries extracted from the Snort ruleset (8,673 patterns) and the ClamAV signature database (5,248 and 2,899 patterns for the type-1 and type-3 sets, respectively). Synthetic input streams were generated by embedding pattern prefixes into a clean corpus – the King James Bible (HTML) for Snort and the concatenation of `/usr/bin` binaries for ClamAV – with controlled match ratios of 1%, 8% and 32%. The reported throughput in MB/s, averaged over 100 runs, shows that FHBM consistently outperforms HBM: gains of approximately +58% and +46% are obtained on Snort and ClamAV type-1 at a 1% match ratio, and +55% on ClamAV type-3 at a 32% match ratio, with peak throughput reaching ≈ 787 MB/s on Snort and exceeding 1,300 MB/s on ClamAV type-3 when the head transition table still

fits within the L3 cache.

Overall, the paper demonstrates that a flexible, state-count-based partitioning of the AC-DFA, combined with SIMD body evaluation and multi-core data parallelism, achieves good throughput on commodity hardware. It also identifies cache capacity as the dominant performance ceiling: once the active state memory exceeds the L3 budget, additional states no longer improve throughput, a finding that reinforces the cache-locality concerns inherent in any shared-memory parallel implementation of Aho-Corasick. Two architectural choices, however, contrast with the present POSIX Threads approach: the authors do not explicitly describe how the input stream is partitioned across the four worker threads nor how cross-boundary matches are handled, and their speedup is tightly coupled to platform-specific SIMD intrinsics rather than to portable data-parallel chunking with overlap. This positions FHBM as a complementary, automaton-restructuring strategy whose throughput gains stem from microarchitectural exploitation rather than from explicit input partitioning, providing a useful comparative reference point for the present Pthreads-based design.

3.3.3 An Optimized Parallel Failure-less Aho-Corasick Algorithm for DNA Sequence Matching

Thambawita et al. [5] investigate the throughput limitations of the canonical Aho-Corasick (AC) algorithm in bioinformatics workloads, where the dictionary may contain thousands of DNA patterns and the input may reach hundreds of megabytes per scan. The authors observe that, although the Parallel Failure-less Aho-Corasick (PFAC) library of Lin et al. already exposes massive thread-level parallelism by launching one thread per input character, it remains a domain-agnostic implementation whose 256-entry transition table and dispersed memory layout do not exploit two structural properties of DNA matching: the alphabet contains only four symbols ($\{A, T, C, G\}$) and the matching kernel has strong spatial and temporal locality. As a result, the baseline PFAC implementation underuses memory bandwidth and has poor cache behaviour on the GPU memory hierarchy.

To overcome these limitations, the authors propose an application-specific optimization of PFAC on a NVIDIA Fermi GPGPU with three coordinated changes. First, the AC transition table is reduced from a generic $256 \times N$ matrix to a compact $4 \times N$ table aligned with the DNA alphabet, drastically shrinking the working set. Second, the transition table is mapped onto texture memory to exploit its two-dimensional spatial caching, while the input text is staged into shared memory to exploit temporal locality across the threads of each block. Third, the `nextState` and `matchedPatternId` arrays are merged into a single contiguous one-dimensional array so that consecutive accesses benefit from cache-line reuse. The underlying parallel decomposition is preserved from PFAC: every input character is assigned its own thread, which independently traverses the

failure-less automaton until a mismatch terminates its scan, and matches are recorded in thread-local result slots that require no inter-thread synchronization during the search phase.

The evaluation is conducted on a NVIDIA Tesla C2075 (Fermi architecture), using five synthetic DNA dictionaries (ranging from 1,000 to 5,000 patterns) and five synthetic DNA input datasets sized between 76 MB and 380.2 MB. The reported metric is wall-clock execution time, measured against the unmodified PFAC library; absolute throughput in MB/s and any sequential CPU baseline are absent. Across all configurations, the optimized implementation consistently outperforms the original PFAC, with the largest datasets exhibiting approximately a $3\times$ speedup over the baseline library. The authors further observe that memory-layout optimizations – in particular the merged one-dimensional array and the use of texture memory – have a substantially larger impact on performance than tuning L1 cache sizes or per-block thread configurations.

This study reinforces two principles that are directly transferable to the present Pthreads-based design even though the target hardware differs: first, that the read-only nature of the AC automaton after construction enables fully lock-free, thread-local matching, irrespective of whether parallelism is realized on a GPU or on a multi-core CPU; second, that the placement of the automaton and the input within the cache hierarchy often matters more than raw thread-count scaling. Two limitations, however, restrict the direct applicability of these findings to the current work. The PFAC decomposition launches one thread per input byte, which is well suited to GPU SIMT execution but mismatched with the much coarser granularity of multi-core CPUs, where the Pthreads model favours chunk-based data partitioning with overlap regions of `max_pattern_len - 1` bytes. Moreover, the alphabet-specific reduction to a $4 \times N$ transition table relies on the small DNA alphabet and cannot be reused for the full 256-symbol byte alphabet of IDS/YARA workloads targeted by the present implementation. The Thambawita et al. study thus serves as a complementary point of comparison, framing PFAC-style fine-grained parallelism as the GPU counterpart of the coarse-grained, chunked Pthreads strategy adopted in this thesis.

3.3.4 The Diversification and Enhancement of an IDS Scheme for the Cybersecurity Needs of Modern Supply Chains

Deyannis et al. [17] examine the performance and energy cost of software Aho-Corasick matching when an industrial supply chain node must concurrently execute its primary business workload and inspect real-time network traffic. The authors observe that the canonical Snort IDS, although well-tuned, saturates the CPUs of low-power supply chain assets and therefore introduces a deployment gap: edge nodes either drop packets under load or shift inspection costs upstream, which conflicts with the distributed perimeter defence recommended by recent supply chain cybersecurity

frameworks. The bottleneck is clearly in the multi-pattern matching kernel, which Snort implements with Aho-Corasick.

To restore line-rate operation without altering the IDS configuration model, the authors offload the AC engine to the FPGA fabric of a low-cost PYNQ Z1 board (Xilinx Zynq-7000 MPSoC). The fully expanded, failure-free DFA is serialized as a flat two-dimensional integer array indexed by `dfa[state * 256 + char]`, eliminating pointer-chasing and turning every transition into a single arithmetic lookup – a layout structurally identical to the read-only `goto_tbl` adopted by the present Pthreads implementation. The transition table is then fed into a four-stage hardware pipeline implemented in Vivado HLS and clocked at 410 MHz, while the official Snort software stack continues to run on the co-located ARM Cortex-A9 cores; only the AC matching kernel is replaced. Because the design consumes less than 10% of the available LUTs, the authors note that two or more copies of the IP module could be instantiated to expose additional spatial parallelism, projecting theoretical $2.8\times$ and $5.6\times$ accelerations that, however, are not empirically validated.

The evaluation employs four publicly available IIoT traffic datasets – the Malware Traffic Analysis blog capture, DARPA-99, the UNSW ToN_IoT dataset and the NCC Group honeypot trace – replayed against the same Snort ruleset, comparing the FPGA-accelerated build against a fully optimized software Snort baseline running on the identical PYNQ board. The reconfigurable design delivers $1.2\times$ to $1.4\times$ acceleration in pure execution time (e.g., 79.4 s versus 108.9 s on the DARPA-99 trace) and up to $1.4\times$ higher packet-per-second rates on traces dominated by large packets, while underperforming the software baseline by approximately 10% on small-packet traces because of interrupt and DMA overhead. The total board power draw of ≈ 13.8 W is much lower than the ≈ 118 W of a high-end single-socket server; combined with the up-to- $1.4\times$ acceleration, the authors report an energy consumption up to almost $12\times$ lower than that server. Aggregate throughput in Gbps and the exact size of the loaded ruleset are not disclosed.

In sum, the paper confirms that the matching phase of Aho-Corasick is the dominant cost of signature-based IDS in resource-constrained settings and demonstrates that a hardware-pipelined DFA can recover throughput at a fraction of the energy budget. Three observations transfer directly to the present Pthreads-based design: the use of a fully expanded failure-free DFA stored as a flat $N \times 256$ integer array is equally appropriate for shared-memory CPU traversal, since it preserves $O(1)$ per-character transitions while permitting concurrent read-only access by multiple worker threads; the matching phase rather than the automaton construction dominates the cost, justifying the architectural decision to parallelize only the search loop; and packet-size or input-size sensitivity is a workload effect that any chunk-based parallel implementation should evaluate, since small inputs amortize less of the per-chunk start-up cost. The

principal divergence is architectural: while the paper realizes parallelism through hardware pipelining and replicated FPGA IP cores, the present Pthreads design pursues coarse-grained data parallelism through chunked partitioning of the input with overlap regions of $\text{max_pattern_len} - 1$ bytes, exploiting commodity multi-core CPUs and avoiding the reconfigurable-hardware deployment cost.

3.3.5 Fast String Searching on PISA

Jepsen et al. [18] target the bandwidth and per-byte compute bottlenecks that arise in disaggregated data centres, where storage nodes carry weak CPUs and every byte of incoming data must traverse PCI-Express to reach a general-purpose processor. The authors observe that pre-filtering large volumes of stored or in-flight data with multi-pattern string matchers is essential to keep downstream analytics manageable, but that an x86 host inspecting every byte does not scale to disk-array or line-rate workloads. Aho-Corasick is the standard algorithm for this pre-filter, and the paper poses the question of whether its deterministic finite automaton can be executed entirely within the data plane of a programmable network ASIC, exploiting native pipeline parallelism rather than thread-level parallelism on a CPU.

To answer this question, the authors introduce PPS, a P4-based compiler and runtime that maps an Aho-Corasick-derived DFA to the PISA architecture of the Barefoot Tofino switch. Two transformations are applied to the canonical automaton. First, a k -stride DFA is created to process k bytes at a time, so that each pipeline pass consumes several characters at the cost of a multiplicatively larger transition table; this trades on-chip SRAM for higher per-stage throughput. Second, the resulting DFA is optionally partitioned into up to three independent sub-automata that are placed in different pipeline stages and traversed in parallel against the same packet, allowing the aggregate pattern set to exceed the memory of any single stage. To enforce correctness when packets are processed independently, the system mandates that adjacent input chunks overlap by the length of the longest pattern – a constraint conceptually identical to the $\text{max_pattern_len} - 1$ overlap adopted by the present Pthreads implementation, although applied here at the network MTU granularity. An optional CRC16 hash representation of the transition table is offered as an accuracy-for-memory trade-off; this approximation is incompatible with exact-match requirements and is therefore restricted to use cases tolerant of false positives.

The evaluation contrasts PPS running on a Tofino switch against two CPU baselines – Spark SQL on a four-node dual-socket Intel Xeon E5-2603 cluster and Unix `grep` on a 12 GB RAM-disk log – using Snort community-rule `content` patterns of up to 32 characters and three corpora: the 4.7 GB Podesta e-mail archive, a 212 GB Twitter streaming capture, and the synthetic log file. End-to-end execution times measured at 128 patterns yield a $> 20\times$ speedup over Spark SQL on the e-mail workload and a

6.5 $\times$ speedup on the tweet workload (5.43 min versus 35.4 min), while throughput at 100 Snort patterns with stride 4 is calculated – not measured – at 3.8 Tb/s for a 64-port Tofino. The authors are explicit that the Tb/s figure is a theoretical upper bound derived from the deterministic line rate of the compiled P4 program; CPU baselines are measured end-to-end, and external GPU/FPGA references (e.g., 150 Gbps on GPU, 45 Gbps for DFC on x86) are quoted rather than reproduced.

The paper shows that Aho-Corasick is an algorithm that scales from a single CPU core all the way to multi-Tb/s switching silicon, and it makes two design decisions that map directly onto the present Pthreads-based work. The mandatory overlap of *longest-pattern* bytes between adjacent input chunks is the same correctness primitive that the present implementation enforces by reserving `max_pattern_len - 1` warm-up bytes at every chunk boundary; this independent confirmation is particularly valuable because it derives from an entirely different parallel substrate, showing that the overlap is not just an artifact of thread-level parallelism. Likewise, DFA partitioning provides a precedent for splitting a pattern set across independent execution units, although the present work partitions the *input* rather than the automaton, leaving the goto table intact for shared, read-only access by all Pthreads workers. The principal divergence is the hardware target: PPS realizes parallelism through replicated pipeline stages and TCAM/SRAM primitives in a switch ASIC, whereas the present design pursues coarse-grained data parallelism on commodity multi-core CPUs, accepting lower theoretical throughput in exchange for portability and exact matching.

3.3.6 Practical Multi-Pattern Matching Approach for Fast and Scalable Log Abstraction

Tovarňák [19] addresses the throughput limitations of multi-pattern matching in log-abstraction pipelines, where each incoming log line must be classified by matching it against a potentially large set of regex-like patterns. The author observes that the naive strategy of iterating over the entire pattern set per log line incurs $O(|\text{patterns}|)$ cost per input and degrades severely once the dictionary contains more than a few tens of entries, while general-purpose multi-regex libraries such as Google RE2 hit memory-cap limits when the DFA representation exceeds a few hundred patterns. The work is positioned alongside Aho-Corasick as a multi-pattern matcher built on a trie-based automaton, but it explicitly targets the same parallelism model adopted by the present thesis: a single automaton constructed once and shared, read-only, across multiple concurrent workers on a commodity multi-core CPU.

To recover throughput, the author proposes REtrie, a compressed trie that combines literal-character pointers with regex *token* pointers (named wildcards) and supports controlled backtracking only over the token segments, preserving an $O(|\text{input}|)$ matching cost in the typical case. Three design choices map directly onto the present

Pthreads design: the trie is constructed sequentially and then declared immutable, so that worker processes need no synchronization during the matching phase; node arrays are compressed with PATRICIA-style path compression and two-level adaptive nesting, lowering the memory footprint of the otherwise sparse 256-entry pointer arrays and improving cache locality; and data-level parallelism is realized by routing disjoint subsets of input log lines to independent Erlang OTP processes, each of which traverses the same shared automaton without any inter-process locking. The system is implemented in Erlang with native-code generation via the HiPE compiler, so the parallelism model is process-based rather than thread-based, but the architectural invariant – read-only shared automaton, thread-local results, no locks on the hot path – is the same one adopted by the Pthreads design of the present thesis.

The evaluation is conducted on a dual-socket Intel Xeon E5-2650 v2 (2.60 GHz) with 64 GB of RAM, scaled from one to eight cores. The pattern sets include a hand-curated set of 22 syslog `auth/authpriv` patterns (P_{22}) and large Apache Hadoop sets (R_x) with up to 3,500 entries derived from source-code analysis. The runtime workloads include D_0 , a real-world capture of 100 million syslog lines collected over one month, plus partially synthetic derivatives (D_x, H_x) of 1.1 million messages each constructed by controlled duplication and shuffling. Each measurement is averaged over ten runs with extreme values discarded. The single-core comparison shows that REtrie's running time remains essentially constant as the pattern count grows from 1 to 22, whereas the naive baseline reaches roughly 12 s on P_{22} versus ≈ 3 s for REtrie and RE2; for $R_x \geq 2,000$ patterns the Erlang REtrie even surpasses C++ RE2, which collides with its default DFA memory limit. The multi-core scaling on D_0/P_{22} exhibits a near-ideal $2\times$ speedup at two cores; at eight cores the 100-million-line workload is processed in 52.543 s, equivalent to $\approx 1,903,203$ log lines per second, with the speedup growth flattening past four cores, in line with the ceiling expected from Amdahl's law [20] and from memory bandwidth.

Taken together, the paper provides independent empirical validation of two design choices that underpin the present Pthreads-based implementation of Aho-Corasick. First, the architectural invariant of a read-only shared automaton accessed concurrently by multiple worker processes is shown to deliver near-ideal scaling at low core counts on commodity multi-core hardware, with the diminishing-returns regime located around four to five cores, an observation consistent with the early plateau found in the present experiments. Second, the use of path compression and adaptive node sizing to improve cache locality reinforces the importance of the flat `goto_tbl` layout employed by the present implementation, even though the algorithmic context differs. Two limitations restrict the direct transferability of the results. The REtrie automaton is not an Aho-Corasick DFA: it lacks the failure and dictionary-suffix functions and processes each log line as a single bounded record, so the chunk-boundary overlap problem that motivates

the present implementation's `max_pattern_len - 1` warm-up region simply does not arise here. Moreover, Erlang OTP scheduling abstracts away the cache, false-sharing and NUMA effects that the Pthreads design must address explicitly. The Tovarňák study therefore serves as a precedent for the parallel *strategy* (lock-free shared immutable automaton) rather than for its concrete realization, and as a useful throughput reference point on comparable Xeon-class hardware for multi-pattern matching at the line-of-text granularity.

3.3.7 Pattern Matching in YARA: Improved Aho-Corasick Algorithm

Regéciová et al. [21] address a particular failure mode of the canonical Aho-Corasick implementation embedded in the YARA rule-matching framework, a representative production scanning tool that relies on this algorithm. The authors observe that YARA's preprocessing phase extracts short literal substrings – termed *atoms* – to seed the AC automaton, but the atom-selection heuristic fails to represent the wildcards, character classes and metacharacters that increasingly appear in industrial signatures. As a result, the scanner either selects empty atoms, in which case the matching loop degenerates into a naive byte-by-byte comparison, or it issues a “slowing down” warning that renders the rule unusable on large-scale platforms such as VirusTotal Hunting. The bottleneck is therefore not in raw matching throughput but in the construction-time choice that determines which literal fragments seed the AC automaton.

To overcome this limitation, the authors propose MYARA, a fork of YARA 4.0.2 introducing three coordinated changes to the preprocessing phase while leaving the matching loop intact and sequential. First, atoms are encoded as 256-bit *bitmaps*, where each bit indicates whether the corresponding byte value is admissible at that position; a literal character, the dot metacharacter and a class such as `[a-z]` are therefore expressed in a uniform structure that supports fast subset and inclusion tests through bitwise operators. Second, the AC `goto` function is generalized to accept *sets* of characters per transition rather than single bytes, and the resulting non-determinism is resolved by an explicit determinization step before the failure function is computed. Third, the existing Aho-Corasick Interleaved State Matrix (ACISM) – a flat array of 32-bit integers that encodes both `goto` transitions and failure links for $O(1)$ lookup – is updated to accommodate the denser slot occupancy produced by multi-symbol transitions. Because the resulting automaton remains entirely read-only after construction, the same shared-memory invariant relied upon by the present Pthreads design continues to hold.

The evaluation is conducted on a server equipped with an Intel Xeon E5-2697 v4 at 2.30 GHz running Debian 4.9.168, with binaries built using GCC 8.3.0. Three pattern dictionaries of contrasting composition are used: a hand-crafted set of five YARA rules targeting Bitcoin addresses, e-mail addresses and the INI format; the publicly available ReversingLabs corpus comprising 371 rules with 0% regular expressions; and the

proprietary Avast set of 13,026 rules, of which only 2.1% are regular expressions. All workloads are scanned against an 8.54 GB real-world corpus that aggregates Bitcoin transaction history, e-mail archives and INI configuration files; each reported measurement is the minimum of fifteen daily runs collected over six days. On the degenerate Bitcoin regex (v1) the scan time falls from 169.700 s under canonical YARA to 17.623 s under MYARA, i.e. a $\approx 9.6\times$ speed-up that recovers the rule from the unusable regime. On the realistic ReversingLabs ruleset, where atoms are dominated by literal and hexadecimal strings, scanning time is essentially unchanged (1,503.65 s vs 1,504.90 s); on Avast the global improvement is approximately 1%. Throughput in MB/s and memory consumption are not reported.

On balance, the paper offers two contributions that bear directly on the present Pthreads-based work even though it introduces no parallelism of its own. First, it documents and characterizes the ACISM representation that the present implementation – and any other shared-memory parallel Aho-Corasick variant – can adopt as a strictly read-only substrate, with the additional benefit that bitmap-encoded transitions remain compatible with the flat `goto_tbl[state * 256 + byte]` layout adopted in this thesis. Second, it provides documented single-threaded baselines on a 8.54 GB realistic corpus that serve as external reference points for the present evaluation. Three limitations restrict the direct applicability of the results, however: the optimization is concentrated in the construction phase, whereas the thesis specifically targets the matching phase; the reported gains evaporate on rulesets dominated by literal patterns (the ReversingLabs case), which closely resemble the IDS and YARA workloads exercised in the present evaluation; and the absence of throughput measurements in MB/s precludes a direct quantitative comparison. The Regéciová et al. study therefore complements rather than competes with the present work: it improves the automaton fed to the scanner, while this thesis improves the parallel execution of the scanner against an unchanged automaton.

3.3.8 SIMD-Accelerated Regular Expression Matching

Sitaridi, Polychroniou and Ross [22] focus on the sequential bottleneck that arises when a DFA-based pattern matcher is embedded in an in-memory database column-filter operator: the per-character transition $s_{t+1} = \delta(s_t, c_t)$ creates a tight data dependency that defeats straightforward SIMD auto-vectorization, leaving most lanes of a wide vector register idle. The authors explicitly note that Aho-Corasick is equivalent to a minimal DFA once the failure transitions are absorbed into the transition table, so the analysis of vectorized DFA traversal performance transfers directly to the matching phase of any AC engine. Their focus is therefore the same matching-phase throughput targeted by the present thesis, but addressed at a different level of the parallelism hierarchy.

To exploit the unused lanes, the paper introduces a non-lockstep, short-circuit SIMD-DFA traversal algorithm. Rather than advancing all W lanes of the SIMD register in unison one character at a time – which wastes work whenever a string is rejected or accepted early – the algorithm tracks an independent input offset and DFA state per lane, uses SIMD *gather* instructions to fetch the relevant byte for each lane in a single instruction, and replaces any lane that reaches a sink state with a fresh input through selective loads. Three auxiliary techniques sustain the throughput: an eight-byte buffered gather amortizes the cost of non-contiguous loads, a two-way unrolled inner loop hides instruction latency, and a branch-free selective store records the identifiers of accepted strings. Critically for the present discussion, the authors compose this SIMD-level data parallelism with thread-level parallelism by spawning one worker per hardware thread; the DFA transition table is shared, read-only, across all workers, and string offsets are partitioned among them. This is structurally identical to the present Pthreads design, with the difference that the thesis partitions a *single* large input into chunks, whereas the paper partitions a column of many short strings.

The evaluation contrasts two platforms: an Intel Xeon E3-1275v3 with four Haswell cores and 256-bit AVX2, and an Intel Xeon Phi 7120P with 61 cores and 512-bit SIMD. Pattern dictionaries include a 90-state URL-validation regex (23 KB transition table), a 9-state e-mail regex, and synthetic multi-pattern dictionaries built from random English words whose DFA size is swept from 10 to 10^5 states to cross the L1, L2 and last-level cache boundaries. Inputs are synthetically generated string columns of varying length, selectivity and average failure point. Throughput is reported in GB/s of total bytes scanned together with the fraction of peak DRAM bandwidth achieved. The non-lockstep algorithm yields up to $2\times$ over a scalar baseline on the Haswell CPU and up to $5\times$ on the Xeon Phi, with the gap over lockstep reaching $3\times$ for long strings that reject early. Two further observations are particularly relevant. First, throughput scales linearly with hardware-thread count on both platforms, confirming that SIMD-level and thread-level parallelism compose without contention as long as the DFA is shared read-only. Second, performance peaks when the DFA fits in the L1/L2 cache: once the transition table grows past last-level cache, vectorization gains shrink, with throughput reverting to a memory-bandwidth-bound regime.

Ultimately, three findings carry directly into the present Pthreads-based work, with the caveat that the underlying parallelism mechanisms differ. First, the explicit observation that AC is equivalent to a minimal DFA, combined with the demonstrated linear scaling of thread-level parallelism over a shared read-only transition table, independently validates the architectural choice of immutable post-construction `goto_tbl` central to the present design. Second, the empirical demonstration that cache residency dominates throughput once the DFA exceeds last-level cache provides a documented mechanism behind the saturation that the present thesis observes on large rule sets,

motivating both the cache-locality concerns of the flat `goto_tbl[state * 256 + byte]` layout and the evaluation of memory-bandwidth contention as the thread count grows. Third, the orthogonality of SIMD-level and thread-level parallelism documented in this paper makes it a natural reference for positioning the Pthreads strategy as a baseline that future work could combine with SIMD intrinsics in the inner loop. Two limitations restrict the transferability of the numerical results: the workload is restricted to fixed-length strings in database columns, in contrast to the variable-length chunked file scans targeted by the present thesis, and the cache-residency assumption holds only for the small URL regex used in most experiments and breaks down for the large multi-pattern dictionaries that motivate AC in security applications.

3.3.9 Harry: A Scalable SIMD-based Multi-literal Pattern Matching Engine for Deep Packet Inspection

Xu et al. [23] tackle the per-byte cost of multi-literal pattern matching in software Deep Packet Inspection (DPI), which the authors identify as the dominant bottleneck of production stacks such as Snort, Suricata, ModSecurity and CloudFlare’s WAF. The canonical Aho-Corasick (AC) algorithm is acknowledged as the traditional choice but is shown to leave large performance margins on the table compared with the SIMD-accelerated FDR engine of Hyperscan; FDR itself, however, requires three SIMD operations per input character regardless of the vector width, attains only 50% SIMD-lane utilization, and depends on the `VPSLLDQ` shift instruction whose 128-bit lane boundary makes AVX-512 unusable. The paper therefore poses the question of how far the throughput ceiling of single-threaded multi-literal matching can be raised on commodity x86 CPUs by aggressively redesigning the SIMD kernel, establishing a useful upper bound against which the multi-core Pthreads strategy of the present thesis can be positioned.

Three coordinated techniques constitute the Harry engine. A column-vector-based shift-or matching algorithm replaces FDR’s row-vector decomposition: instead of loading one row per input character, Harry loads one column vector per literal-position slot, so that the number of SIMD operations becomes proportional to the literal length ℓ (fixed at eight in the implementation) rather than to the input length, yielding only 0.41–0.71 operations per input character on AVX-512 and raising vector utilization to 87.5%. Two mask-table encodings – a compression-based variant (*Harry6b*, optimal for small rule sets) and a decomposition-based variant (*Harry12b*, optimal for large rule sets) – are selected at compile time by a heuristic based on the zero-bit rate of each encoding, keeping the column vector within 512 bits in both regimes. Finally, the legacy `VPSLLDQ` shift is replaced with a `VPERMB`-based shuffle, eliminating cross-lane data loss and enabling the kernel to exploit AVX-512 without algorithmic regressions. The matching pipeline preserves Hyperscan’s two-stage structure – approximate filter followed by

exact verification – so the read-only nature of the underlying tables remains intact, consistent with the lock-free shared-automaton invariant adopted by the present Pthreads design.

Harry is evaluated on an Intel Xeon Gold 6348 (2.60 GHz, two sockets of 28 cores each, AVX-512) running Ubuntu 20.04, with GCC 9.4.0 used without optimization flags. Literal rules are extracted from two production rule sets – OWASP ModSecurity Core Rule Set v3.3.2 and Snort Emerging Threats v2.9.0 – and partitioned into nine subsets spanning 10 to 3,000 rules. Three input traces are used: 121 MB of real IXIA HTTP packets, 4 MB of Alexa non-HTTP packets and 763 KB of synthetic random bytes; each experiment is averaged over 1,000 runs. The reported throughput ranges from 30 to 70 Gbit/s and represents a $26\times$ – $45\times$ speed-up over canonical Aho-Corasick on real HTTP and non-HTTP traffic, peaking at $52\times$ on the random byte stream, and a $1.06\times$ – $2.09\times$ improvement over FDR. The heuristic encoding selector picks the better variant in every evaluated configuration, and Harry maintains its advantage across the full 10–3,000 rule range, in contrast to the prior Teddy engine, which degrades sharply beyond roughly sixty rules.

Overall, the paper has two effects on the related-work positioning of the present thesis. First, it concretely quantifies the ceiling that single-threaded SIMD literal matching can reach on commodity AVX-512 hardware – 30–70 Gbit/s at 0.41 operations per character with 87.5% vector utilization – and reports that this regime is up to $52\times$ faster than the canonical Aho-Corasick algorithm. This establishes Harry as the contrast point against which a coarse-grained Pthreads parallelization of AC must justify itself, and clarifies that the rationale of the present work is not to outperform Harry on a single thread but to exploit multi-core resources where the SIMD path is already saturated, since the same input can be partitioned across cores and the AC automaton kept fully shared and read-only. Second, the paper documents that SIMD-level and thread-level parallelism are architecturally orthogonal: Harry runs entirely within a single thread, so its gains compose multiplicatively with any data-parallel chunking strategy on multi-core CPUs. Two caveats restrict the direct numerical comparability of the results: the AC baseline is compiled by GCC without optimization flags while the Harry path is hand-vectorized, which likely inflates the reported $52\times$ ratio, and the engine replaces rather than parallelizes the Aho-Corasick automaton, departing from the strict-AC requirement of the present thesis. Harry should therefore be cited as the canonical ceiling for single-threaded literal matching in DPI, not as a parallel AC variant.

3.3.10 Characterizing Realistic Signature-based Intrusion Detection Benchmarks

Aldwairi, Alshboul and Seyam [6] address a methodological gap that affects every empirical study of pattern matching for signature-based Intrusion Detection Systems (NIDS), including the present thesis: there is no agreed standard for the datasets, the

metrics or the simulation environment under which Aho-Corasick and competing algorithms are evaluated, and the resulting heterogeneity makes cross-paper comparisons unreliable. The authors review the most cited public datasets – DARPA, KDD Cup 99, NSL-KDD, Kyoto 2006+ and CAIDA – and document that all of them were designed for anomaly-based detection and therefore lack representative attack signatures; privately collected traces, conversely, cannot be redistributed for legal reasons. The paper therefore positions itself not as a parallelization contribution but as the construction of a reproducible NIDS benchmark stack that subsequent works – this thesis included – can adopt to ground throughput and speed-up claims in realistic conditions.

The contribution is delivered as a five-part artifact. First, a GUI-based parser ingests heterogeneous IDS rule formats and extracts the `content` and `uricontent` keywords of 120 Snort rule files, yielding hexadecimal-encoded signature dictionaries of varying cardinality (from 300 to over 2,400 signatures, suitable for scalability sweeps). Second, eight representative traces are curated from the 78 DEFCON17 Capture-the-Flag captures: four “good” traces (audio-video streaming, downloads, Hotmail, live chat) representing baseline traffic, two “ugly” traces (51 and 58) where 45% and 43% of packets carry attack signatures, and two “bad” traces (1 and 22) with intermediate 16%–17% malicious densities. Third, a pluggable matching engine implemented as a Windows DLL allows researchers to substitute their own algorithm under identical input and instrumentation. Fourth, four classical algorithms – Aho-Corasick, Knuth-Morris-Pratt, Wu-Manber and Rabin-Karp – are reimplemented strictly sequentially against the same engine to provide single-thread reference numbers. Fifth, an explicit set of metrics is fixed: runtime in microseconds, memory in bytes, and count of matched attack signatures per trace, each averaged over ten repetitions.

The hardware platform is not specified; the engine is implemented in C# and the authors flag that the runtime’s garbage collector limits the precision of memory measurements. The reported numbers are nevertheless useful as a sequential baseline. Aho-Corasick is the slowest of the three viable algorithms on the large “audio-video” trace, taking 2,374,992 μ s against Wu-Manber’s 619,469 μ s, and consumes the largest memory footprint, peaking at 6,840,694 bytes versus 2,694,016 bytes for KMP. The runtime-versus-signature-count sweep on the same trace shows that AC scales non-linearly: the time grows from 168,079 μ s at 300 signatures to 2,374,992 μ s at the full dictionary, an effect the authors describe as “almost exponential”. The memory-versus-packet-count sweep on the “ugly” trace 58 reports a similar growth – from 29,402 bytes at 25,000 packets to 3,871,408 bytes for the full trace – although this expansion most likely reflects accumulated match-record storage and C# allocation behavior rather than an intrinsic growth of the AC automaton, whose DFA trie is fixed by the signature set.

In sum, the paper provides the present thesis with two contributions that no other reviewed work delivers in combination. First, the DEFCON17 trace selection and the

Snort-derived signature dictionaries are documented, redistributable benchmark inputs that can be replayed against the present Pthreads implementation to expose the multi-core speed-up under the same workload conditions used elsewhere in the literature. Second, the per-trace sequential AC runtimes – in the order of tens of milliseconds for the smaller traces and a few seconds for the largest – furnish numerical reference points against which the present multi-thread throughput can be situated. Three limitations restrict the direct use of the published numbers. The hardware on which they were measured is not specified, so absolute runtimes cannot be reproduced. The implementation language (C#) and its garbage collector make memory figures imprecise. And the paper does not report throughput in MB/s or Gbps, which are the metrics the present thesis adopts; conversion from microseconds plus payload size in bytes is therefore required when quoting the Aldwairi figures as competitors. The paper should be cited for benchmark methodology and realistic-workload selection rather than as a parallel-AC technique, and it complements the multi-core, multi-threaded contributions reviewed above by anchoring the evaluation in the canonical NIDS context that motivates Aho-Corasick parallelization in the first place.



4

4

PROPOSED SOLUTION

This chapter presents the proposed solution: a modular parallel framework for the Aho-Corasick multi-pattern matching algorithm, targeting shared-memory multi-core CPUs. The design is guided by a single structural decision from which every other property follows. The execution is split into separate phases: the master thread builds the automaton sequentially, after which its state-transition tables are declared strictly read-only. Because the automaton is never mutated during scanning, multiple worker threads can traverse it concurrently without any mutual-exclusion primitive on the hot path. This lock-free property is what allows the parallel search to approach the limits imposed by the hardware rather than by software contention.

The framework is organized so that distinct optimization ideas can be plugged in and measured independently. Two complementary forms of parallelism are considered: partitioning the *input text* among threads, and partitioning the *pattern dictionary*. In addition, an algorithmic optimization of the match-emission step is proposed that benefits both the sequential and the parallel cases. The remainder of this chapter describes each component.

4.1 DATA PARTITIONING AND BOUNDARY OVERLAP

To exploit thread-level parallelism on a single large input, the framework partitions the text buffer into contiguous segments, one per worker thread. Each worker traverses the shared automaton over its own segment and records matches in a private list, so that no synchronization is required while scanning.

However, a naive partitioning of the byte stream would miss any pattern that crosses the boundary between two adjacent segments, because neither worker would observe the full pattern. To prevent these boundary false negatives, the design uses a controlled overlap. Let $L_{\max}$ be the length of the longest pattern in the dictionary. A worker responsible for the segment spanning offsets $[S, E)$ does not begin scanning at S but at $S - L_{\max} + 1$, except for the first worker, which starts at offset 0. This warm-up margin guarantees that, by the time the worker reaches the start of its owned region at S , the automaton is in the same state it would have reached in a single sequential pass from offset 0, so every boundary-crossing pattern is observed by exactly one worker. Any match whose end position falls before S is suppressed, because it is owned and

reported by the previous worker. Conversely, assigning each match to the segment containing its end position keeps ownership disjoint. This preserves an exact one-to-one correspondence with the sequential baseline and avoids duplicate detections. An overlap of $L_{\max} - 1$ is the minimum that achieves this property; any smaller value loses matches, and any larger value only adds redundant work.

4.2 LOAD BALANCING ON HETEROGENEOUS CORES

Uniform segments are optimal only when all cores are equally fast. On a hybrid processor, where Performance and Efficiency cores run at different clock frequencies, equal-sized segments cause the threads on slower cores to finish late, leaving the faster threads idle at the final synchronization barrier. The proposed framework therefore includes a topology-aware partitioning strategy that detects the heterogeneous layout and scales each segment in proportion to the relative speed of the core processing it. The goal is to equalize the finishing times of all workers, reducing the idle time wasted at the barrier. As an alternative that requires no knowledge of the topology, the framework also supports dynamic dispatch, in which the text is divided into many small tasks that idle workers claim through a single atomic counter, so that faster cores naturally process more tasks.

4.3 FLATTENING THE MATCH-EMISSION PATH

In the canonical Aho-Corasick algorithm, each accepting state references the patterns that end there through linked structures, and reporting all matches at a given state requires walking these pointer chains. With a large dictionary, these pointers are scattered across an automaton much larger than the cache. As a result, each emission causes cache misses and serialized pointer dereferences that compete with the hot state-transition traffic.

To remove this cost, the design proposes a flattened output layout. During construction, the patterns associated with every state are collected into a single contiguous array, and each state stores only a starting offset and a count into that array. At scan time, emitting the matches of a state becomes a sequential read of a short, contiguous range. The hardware prefetcher handles this efficiently, removing pointer dereferences from the inner loop. This optimization is purely a memory-layout change: it preserves the exact set of reported matches and is inherited by every scanning variant, whether sequential or parallel.

4.4 PARTITIONING THE DICTIONARY

All the strategies above partition the text while keeping the dictionary intact. A complementary approach, evaluated for contrast, partitions the dictionary instead. In this approach, the pattern set is divided into K groups based on the first byte of each pat-

tern, and a smaller sub-automaton is built for each group. Every sub-automaton is then scanned over the input. The motivation is that each sub-automaton has a smaller working set, which might fit better in the cache and reduce the memory pressure of a single monolithic automaton. The framework also supports a two-dimensional composition that combines dictionary sharding with text chunking. Whether either approach actually outperforms plain text parallelism on memory-bound workloads is an empirical question, answered in the results chapter.



5

5

METHODOLOGY

This chapter describes the experimental methodology adopted to design and evaluate the parallel scanning strategies introduced in the previous chapter. The central object of study is not a particular processor, but a *computational regime*: the behaviour of multi-pattern matching when the automaton footprint greatly exceeds the last-level cache (LLC), which is the operating point of signature-based intrusion detection over large rule sets. The reference machine is therefore treated as a *measurement platform* rather than as the subject of the investigation, and the conclusions are framed in terms of relative scalability behaviour (speedup, throughput, and efficiency) instead of absolute production capacity.

The methodology is organized around four principles: (i) every parallel variant is validated for correctness against the sequential baseline before any performance claim is made; (ii) each optimization axis is isolated so that its individual contribution can be attributed unambiguously; (iii) the experimental design includes hypotheses that the experiments are intended to *falsify*, not merely to confirm; and (iv) measurement noise inherent to the hardware is reported explicitly rather than hidden.

5.1 EXPERIMENTAL ENVIRONMENT

All measurements were performed on a single Intel Core i5-1235U processor, a hybrid (heterogeneous) design of the Alder Lake generation. The processor integrates two Performance cores (P-cores) with simultaneous multithreading and eight Efficiency cores (E-cores) without it, exposing twelve logical hardware threads in total. The two core types differ in maximum clock frequency (approximately 4.4 GHz for P-cores and 3.3 GHz for E-cores), which is the source of the load-balancing problem addressed by the frequency-weighted partitioning strategy. The memory hierarchy comprises a private L1 data cache per core (48 KiB on P-cores, 32 KiB on E-cores), 1.25 MiB of L2 per P-core (and 2 MiB per E-core cluster), and a single 12 MiB L3 cache shared by all cores. Main memory is 31 GiB of dual-channel DDR4, with a peak bandwidth of approximately 25 GB/s. This 12 MiB shared LLC is the architectural threshold that separates the cache-resident regime from the memory-bound regime studied in this work.

The software stack is a purpose-built laboratory written in C11, compiled with GCC 13.3.0

using `-O3 -march=native` and an aggressive warning profile, and linked against the POSIX Threads library [24]. Experiments ran under Linux (kernel 6.17.0). To minimize scheduling and frequency-scaling artifacts, the CPU frequency governor was set to `performance` for the canonical measurement campaign; topology-aware searchers (`pthread_chunked_v3` and `pthread_chunked_v3_flat`) additionally pin each worker thread to a dedicated logical CPU via `pthread_setaffinity_np`, while the remaining searchers rely on OS scheduling without explicit binding. The build was additionally validated under sanitizer instrumentation, namely the Address and Undefined Behavior sanitizers and the Thread Sanitizer, as discussed in Section 5.7.

5.2 PATTERN DICTIONARIES AND TEXT CORPORA

The workloads were chosen to sweep the automaton footprint across the cache hierarchy while keeping the input text realistic for the intrusion-detection domain. Table 2 summarizes the pattern dictionaries, which were derived from publicly available Snort and Emerging Threats rule sets, and Table 3 summarizes the text corpora.

Table 2: Pattern dictionaries and the resulting automaton footprint.

Dictionary	Patterns	States	Automaton	Cache regime
Snort-100	100	1,939	1.93 MiB	Well within L3 (cache-resident)
Snort-1k	~1,000	11,974	11.92 MiB	Near the L3 boundary
Snort (full)	4,188	55,479	55.23 MiB	Exceeds L3 (moderate)
Emerging Threats	44,678	508,896	506.78 MiB	42× the L3 (memory-bound)

Source: Prepared by the author from Snort and Emerging Threats rule sets.

Table 3: Text corpora used as scanning input.

Corpus	Size	Characterization and role
SimpleWiki	~1.19 GiB	Natural-language encyclopedic text with low match density; used as the contrast corpus in the cross-corpus experiment, which isolates the effect of the text content.
Enron	~1.32 GiB	Real e-mail messages with medium match density; the canonical benchmark corpus.
Enron (8×)	~10.59 GiB	The Enron corpus replicated eightfold; amortizes the build phase and reduces scheduler and thermal noise in scaling curves.

Source: Prepared by the author. The Enron corpus derives from the public Enron Email Dataset.

The two smaller dictionaries (Snort-100 and Snort-1k) serve as *contrast regimes*: their automata fit in or near the cache and expose the cross-over point at which memory pressure begins to dominate. The full Snort dictionary and the Emerging Threats

subset represent the actual target of the study, the latter being the most aggressive architectural stressor, with an automaton roughly forty-two times larger than the shared LLC.

5.3 EXECUTION MODEL AND INVARIANTS

Every execution of the laboratory follows three strictly separated phases; this separation is the foundation of the lock-free design.

1. **Construction.** The patterns are inserted into a trie, and a breadth-first traversal computes the goto, failure, and dictionary-suffix links, together with the precomputed output structures. The resulting automaton is then declared immutable.
2. **Search.** Each worker thread reads the shared, read-only automaton and its assigned portion of the input text, emitting matches into a private, thread-local list. No mutex or atomic operation is used on the hot path, because the automaton is never modified during this phase.
3. **Merge.** After all workers join, the master thread concatenates the thread-local lists in a deterministic order. This phase is always sequential and accounts for less than 1% of the total time, so it is not a target for optimization.

This study respects a set of non-negotiable invariants that make the parallelization sound. The automaton is strictly read-only during the search; the hot loop uses neither locks nor atomics; matches remain thread-local until the join; and the partitioning preserves match ownership without duplication. For any text-level partitioning, the minimum overlap between adjacent segments is $L_{\max} - 1$ bytes, where $L_{\max}$ is the length of the longest pattern — as established in the previous chapter, the smallest margin that guarantees that a boundary-straddling pattern is detected by exactly one worker. Match ownership is made disjoint by assigning each boundary-crossing match to a single segment according to its end position.

5.4 EVALUATED STRATEGIES

To attribute each performance effect to a single cause, the strategies were organized as an additive grid over four orthogonal axes, all selectable at runtime within the same binary and all sharing the same automaton. Beyond the sequential baseline, the controlled set of variants is summarized in Table 4.

The text-parallelism axis partitions the input across threads; the output-layout axis addresses the cost of emitting matches; the dictionary axis partitions the pattern set instead of the text; and the composition axis tests whether the text and dictionary strategies combine. The construction phase has a separate, optional level-synchronous parallel variant. Because the automaton produced by the parallel construction is bit-for-bit

Table 4: Controlled set of evaluated strategies, grouped by optimization axis.

Axis	Strategy
Baseline	Single-threaded sequential scan (reference for correctness and speedup).
Text parallelism	Static homogeneous text chunking with boundary overlap. Refined static chunking with cache padding and a split warm-up/owned loop. Topology-aware chunking that weights segment sizes by core frequency. Dynamic chunk dispatch through an atomic task counter (bag of tasks).
Output layout	Flattened, contiguous output table that replaces pointer-chasing emission.
Dictionary parallelism	Dictionary sharding by pattern prefix into K independent sub-automata.
Composition	Two-dimensional combination of dictionary sharding and text chunking.

Source: Prepared by the author.

identical to the sequential one, any pairing of a construction mode with a search strategy is valid, which cleanly isolates the contribution of each axis.

5.5 METRICS

The primary metrics follow the standard definitions introduced in the theoretical framework. Search throughput is reported in MB/s (and, where useful, in Gbps); speedup is the ratio of sequential to parallel search time as in Eq. 1; and parallel efficiency is the speedup normalized by the thread count as in Eq. 4. The construction time is measured and reported separately from the search time, so that the cost of rebuilding the automaton (relevant to rule reloading) is never conflated with scanning throughput. When a hypothesis concerns load balancing between heterogeneous cores, per-thread timings and the coefficient of variation across repetitions are reported as a direct measure of synchronization waste at the final barrier. Finally, the automaton footprint in bytes is recorded as the causal variable that explains throughput across regimes, rather than as a headline figure in its own right.

5.6 MEASUREMENT PROTOCOL

Each reported data point follows the exact per-phase repetition protocol summarized in Table 5. Phase A (speedup curves) sweeps thread counts $T \in \{1, 2, 4, 6, 8, 10, 12\}$ for the canonical Enron corpus and $T \in \{1, 4, 8, 12\}$ for the eight-fold replica, as recorded in `sweep.db`. Phases A–C (speedup curves, footprint, and cross-corpus) each execute

two warm-up iterations followed by five timed iterations; phase D (per-thread diagnostic) executes one warm-up followed by three timed iterations; and phase E (parallel construction) executes a single timed build with no warm-up. The warm-up iterations pay for page faults and warm the caches so that the timed iterations measure steady-state behaviour. Both the mean and the best (minimum) times are recorded; the mean is used for the reported throughput, and the dispersion between iterations is retained as an indicator of stability.

Table 5: Per-phase repetition protocol of the formal sweep (2026-05-29), verified against `sweep.db`.

Phase	Description	Warm-ups	Timed iters
A	Speedup curves	2	5
B	Footprint sweep	2	5
C	Cross-corpus	2	5
D	Per-thread diagnostic	1	3
E	Parallel construction	0	1

Source: Prepared by the author from `sweep.db` (sweep 2026-05-29).

A caveat specific to the measurement platform must be stated explicitly. The reference machine is a thermally constrained mobile processor. Under sustained all-core load lasting hours, the package throttles its frequency, and the measured multi-threaded throughput settles roughly 15 to 35% below what isolated bursts on a cold processor report. The values published in this work are deliberately the sustained, under-load figures, which are conservative and reproducible; cold-burst peaks are treated as the non-sustainable upper bound and are not used as headline results. This thermal sensitivity is itself a finding consistent with the central thesis: at the scale of interest, performance is governed by physical resource limits rather than by the number of threads.

5.7 CORRECTNESS VALIDATION

Performance is only meaningful once correctness is established, so every parallel variant was required to reproduce the output of the sequential baseline before being benchmarked. Correctness was checked at two levels. First, the suite of automated tests requires every strategy to agree with the sequential baseline on the exact set of matches, across thread counts of $\{1, 2, 3, 4, 7, 8\}$. Second, on the full-scale workloads, the MD5 digest of the ordered match stream produced by each parallel strategy was compared against the digest produced by the sequential baseline; agreement was confirmed up to a thread count of 64 (a $5.3\times$ oversubscription of the hardware), demonstrating that the partitioning logic is robust under heavy oversubscription. The match count was verified to be consistent for each (dictionary, corpus) pair across all strategies. Finally, ThreadSanitizer reported no data races during the parallel search, empirically confirming the read-only-automaton invariant and the absence of unsynchronized

shared writes on the hot path.

5.8 EXPERIMENTAL DESIGN AND HYPOTHESES

The experiments were structured to test a set of explicit hypotheses, several of which anticipate negative or qualified outcomes. The main hypotheses are: that text chunking scales sublinearly on the hybrid processor but suffers from load imbalance when segments are sized uniformly; that the automaton footprint, rather than the corpus content, governs throughput once the working set escapes the cache; that the flattened output layout reduces emission cost in both the sequential and parallel cases; that a level-synchronous parallel construction reduces build time for large dictionaries; that prefix-based dictionary sharding can recover throughput in the memory-bound regime by shrinking the per-shard working set; and that a two-dimensional composition of sharding and chunking could combine cache locality with text parallelism. Each hypothesis maps to a controlled experiment that varies a single axis while holding the others fixed, always comparing against the sequential baseline for headline speedup and against the strongest text-parallel strategy when the question concerns composition. Reporting the strategies that fail to outperform their simpler counterparts is treated as a scientific result in its own right, since it delimits the boundary beyond which plausible optimizations stop paying off on this class of hardware.



6

6

RESULTS AND DISCUSSION

This chapter presents and discusses the empirical results, following the protocol described in the previous chapter. All search and construction figures reported here come from the same sustained measurement campaign; the end-to-end pipeline times evaluated later in this chapter were collected in a dedicated benchmark on the same platform and under the same correctness constraints. Every parallel strategy was first verified to reproduce the exact match set of the sequential baseline. The analysis proceeds in a deliberate order: it first characterizes the bottleneck that motivates the entire study (the relationship between automaton footprint and throughput); it then isolates the contribution of each optimization axis; and it finally reports the cumulative behaviour at scale, including the strategies that did not pay off. Throughout, speedup is measured against the single-threaded sequential baseline, and the reported values are the sustained, under-load figures, which are conservative relative to cold-burst peaks.

6.1 CHARACTERIZING THE BOTTLENECK: FOOTPRINT VERSUS THROUGHPUT

The defining characteristic of the problem is that throughput is governed by the size of the automaton, not by the content of the text. Table 6 sweeps the dictionary from one hundred patterns to roughly forty-five thousand, keeping the corpus fixed, and reports the single-threaded scanning throughput. Isolating the effect with a single thread removes any threading confound, so the degradation can be attributed entirely to the memory hierarchy.

Table 6: Single-threaded throughput as the automaton footprint grows.

Dictionary	Patterns	Automaton	Cache regime	Throughput (MB/s)
Snort-100	100	1.93 MiB	Well within L3	483.1
Snort-1k	~1,000	11.92 MiB	Near L3	337.3
Snort (full)	4,188	55.23 MiB	Exceeds L3	240.7
Emerging Threats	44,678	506.78 MiB	42× the L3	125.6

Source: Prepared by the author. Corpus fixed; single thread.

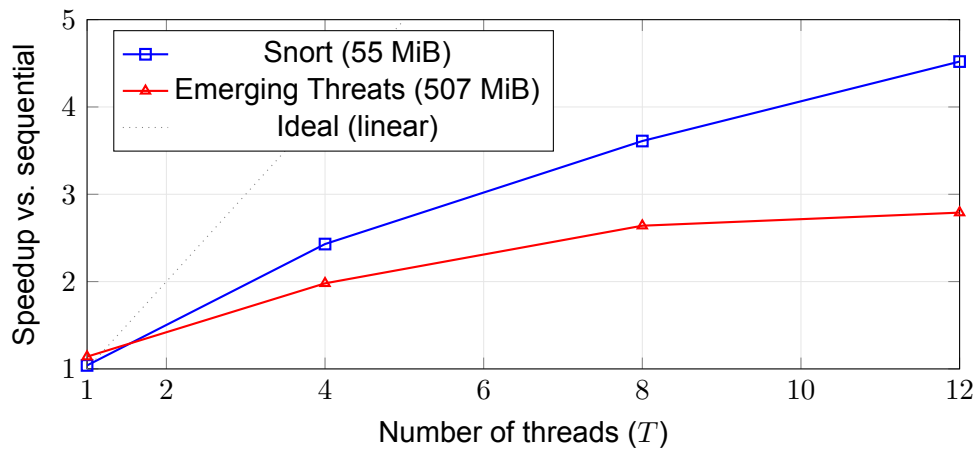
Throughput falls monotonically by 74.0%, from 483.1 MB/s when the automaton fits

comfortably within the shared L3 to 125.6 MB/s when it is forty-two times larger than that same cache. The cause is structural: each input byte triggers a lookup in the state-transition table, and once that table no longer fits in cache, every transition risks a main-memory access. The degradation persists in parallel execution: at twelve threads, the speedup of the same topology-aware chunked strategy drops from $6.68\times$ in the cache-resident regime to $2.63\times$ in the memory-bound regime, confirming that the limit is the memory subsystem rather than a threading artifact. This is the central obstacle to scaling Aho-Corasick at intrusion-detection scale, and it frames the interpretation of every subsequent result.

6.2 SCALABILITY OF TEXT PARALLELISM

The primary parallelization strategy partitions the input text across worker threads. Figure 4 shows the speedup as a function of thread count for the two representative regimes, measured on the largest corpus (~ 10.6 GiB) to minimize scheduler and thermal noise.

Figure 4: Speedup of text parallelism versus thread count, for a moderate dictionary (Snort, 55 MiB; `pthread_chunked_v3_flat`) and a memory-bound dictionary (Emerging Threats, 507 MiB; `pthread_chunked_flat`), on the ~ 10.6 GiB corpus (`enron_x8`).



Source: Prepared by the author.

Both regimes scale sublinearly. With the moderate dictionary, the best text-parallel strategy reaches $4.52\times$ at twelve threads (1,087.1 MB/s; CV 10%); on the canonical corpus, the dynamic-dispatch strategy peaks slightly higher at $4.79\times$ (1,148.1 MB/s; CV 2%), the highest search speedup observed on the intrusion-detection-scale dictionaries. With the memory-bound dictionary, the curve flattens far earlier: the best strategy reaches only $2.79\times$ at twelve threads (CV 8%), and on the canonical corpus the peak occurs near six threads ($2.85\times$; CV 16%) and then *regresses* as more threads are added. This early plateau is the most important observation of the scaling analysis. Once the automaton is forty-two times larger than the cache, the shared DRAM bus saturates, and adding more threads (especially the slower Efficiency cores past the

two Performance cores) no longer increases throughput. The physical limit is memory bandwidth, not the number of cores.

For the moderate-dictionary regime, where memory bandwidth is not the binding constraint, the $4.52\times$ ceiling at twelve threads is consistent with Amdahl's law [9]: the observed 37.7% parallel efficiency implies an inherent serial fraction of approximately 15%—arising from merge coordination and work dispatch—compounded by the P/E asymmetry of the thread pool, in which the four Performance-core hardware threads carry disproportionately more per-thread weight than the eight Efficiency cores.

Parallel efficiency (Eq. 4) quantifies this overhead directly: at twelve threads, the moderate-dictionary regime achieves 37.7% efficiency ($4.52 \times /12$), while the memory-bound regime achieves only 23.3% ($2.79 \times /12$). The gap between the two is caused primarily by DRAM saturation rather than threading overhead, since it mirrors the cache-residency difference and not the per-thread synchronization cost.

A separate experiment fixed the dictionary and strategy while changing the corpus, to confirm that the text content is not the main factor. The sequential throughput on the two corpora is nearly identical (241.0 MB/s versus 248.4 MB/s), and the twelve-thread speedup differs only modestly ($4.42\times$ on the e-mail corpus versus $3.66\times$ on the encyclopedic corpus), a gap within the run-to-run variation of the e-mail measurement ($CV \approx 66\%$). Because Aho-Corasick performs one transition per input byte regardless of match density, the footprint of the automaton dominates the behaviour far more than the characteristics of the corpus.

6.3 THE FLATTENED OUTPUT LAYOUT

The proposed output-layout optimization replaces the pointer-chasing emission of matches with a sequential read of a contiguous array. Because it changes only the memory layout, its effect can be measured in isolation on a single thread. Table 7 shows that the gain is regime-dependent and, importantly, grows precisely as memory pressure increases.

Table 7: Single-threaded gain of the flattened output layout, by regime.

Regime (dictionary)	Automaton	Baseline (MB/s)	Flattened (MB/s)	Gain
Snort-100 ($\ll L3$)	1.93 MiB	483.1	488.1	+1.0%
Snort (full, $> L3$)	55.23 MiB	240.7	252.0	+4.7%
Emerging Threats ($\gg L3$)	506.78 MiB	125.6	142.0	+13.0%

Source: Prepared by the author. Single thread.

When the automaton fits in cache, the optimization provides almost no benefit (+1.0%), because the linked structures are already cache-resident. When the automaton is far larger than the cache, the contiguous layout avoids the scattered misses of the pointer

walk and delivers +13.0% on a single thread, which is the cheapest gain available in the memory-bound regime. In the parallel case, the layout is inherited by the chunked strategies, and the flattened variants lead or tie their pointer-chasing counterparts in both intrusion-detection-scale regimes; the best results — $4.52\times$ for the moderate dictionary at the multi-gigabyte scale and $2.85\times$ at the peak of the memory-bound regime on the canonical corpus — come from strategies that incorporate this layout. It is worth noting that the gain does not compound multiplicatively with text parallelism in the moderate regime: there, the DRAM stalls of the transition table already partially mask the emission cost, so the layout provides a smaller improvement than its single-threaded figure would suggest. The optimization is therefore best described as a cheap, broadly applicable improvement whose payoff is largest exactly where the problem is hardest.

6.4 LOAD BALANCING ON HETEROGENEOUS CORES

On the hybrid processor, uniformly sized text segments create stragglers: the threads assigned to Efficiency cores finish later than those on the faster Performance cores, and the early finishers idle at the barrier. The effect is visible not in the mean throughput but in its stability across repetitions. Table 8 reports, for the canonical workload at twelve threads, the coefficient of variation of the per-iteration timing.

Table 8: Run-to-run stability of the text-parallel strategies (twelve threads).

Strategy	Throughput (MB/s)	Variation (CV)
Static homogeneous chunking	1,064.7	39.9%
Static chunking (cache-padded)	902.0	52.9%
Topology-aware (frequency-weighted)	879.8	1.7%
Dynamic dispatch	864.4	3.3%
Flattened output with chunking	1,007.2	0.4%
Topology-aware with flattened output	994.2	4.3%

Source: Prepared by the author. Canonical workload, twelve threads; dedicated stability diagnostic (three timed iterations under sustained load). The mean throughputs are subject to the run-to-run variance the table quantifies and are not directly comparable with the sustained figures of the scalability analysis.

The static homogeneous strategies exhibit a coefficient of variation between 39.9% and 52.9%, consistent with slower cores intermittently delaying the overall execution. Both solutions proposed in the design reduce this variation: frequency-weighted partitioning reduces it to 1.7% and dynamic dispatch to 3.3%, while the flattened chunking strategy is the most stable of all (down to 0.4%) — indicating that uniform partitioning makes the run-to-run timing sensitive to how the scheduler places the slow-core segments, rather than imposing a fixed penalty. This shows that topology awareness does not make the transition function faster; it removes idle time at the barrier, trading

a small change in mean throughput for a large gain in predictability. This is valuable for reproducibility and worst-case latency, but it does not by itself change the ranking of strategies, because the underlying ceiling remains the memory subsystem.

6.5 PARALLEL CONSTRUCTION OF THE AUTOMATON

The construction phase can also be parallelized, because each level of the breadth-first traversal that computes the failure links depends only on shallower levels and can therefore be processed in parallel with a barrier between levels. The trie insertion remains sequential; only the traversal is parallelized. Table 9 reports the resulting build speedup.

Table 9: Speedup of the parallel automaton construction.

Dictionary	States	Sequential build (ms)	Parallel build
Snort-100	1,939	1.5	Regresses (overhead dominates)
Snort (full)	55,479	48.2	1.19× peak (four threads)
Emerging Threats	508,896	485.2	1.55× peak (eight threads)

Source: Prepared by the author.

Parallel construction only pays off for the large dictionary, where it reduces the build time from 485 to roughly 313 milliseconds (1.55× at eight threads, falling to 1.41× at twelve). For the small dictionaries the per-level spawn-and-join overhead exceeds the work, and the build slows down. This optimization should therefore be understood as an operational benefit, a reduction in rule-reloading latency for systems that frequently update their signatures, rather than as a contribution to scanning throughput; on the multi-gigabyte workloads it accounts for less than 1% of the total time.

6.6 PARTITIONING THE DICTIONARY: A QUALIFIED NEGATIVE RESULT

Partitioning the dictionary, rather than the text, was evaluated as an alternative, assigning patterns to shards by their first byte (prefix sharding). Table 10 compares it against plain text parallelism on the canonical workload.

Table 10: Dictionary sharding compared to text parallelism (Snort, canonical corpus).

Strategy	Throughput (MB/s)	vs. sequential	vs. text parallelism
Dictionary sharding (by prefix)	327.2	1.37×	0.34×
Text chunking with flat output	976.1	4.07×	1.00×

Source: Prepared by the author. Sharding at its best shard count (six); text chunking at twelve threads.

Prefix sharding reaches 1.37× over the baseline, and its gain saturates almost immediately — 1.35× at two shards, within 2% of the six-shard peak, declining afterwards

— which indicates that the benefit comes from improved cache locality within each smaller sub-automaton rather than from actual parallelism. Even at its best it attains only about a third of the throughput of plain text parallelism. The conclusion is that, on this class of hardware, partitioning the text dominates partitioning the dictionary: a real but small effect, not a competitive strategy.

6.7 COMPOSING SHARDING WITH CHUNKING: A FALSIFIED HYPOTHESIS

A natural hypothesis is that combining dictionary sharding (for cache locality) with text chunking (for parallelism) would recover the throughput lost in the memory-bound regime, by making each sub-automaton small enough to approach the cache. Table 11 tests this two-dimensional composition against flat-output text chunking and disproves the hypothesis.

Table 11: Two-dimensional composition versus flat-output text chunking (twelve threads).

Workload	Composition (MB/s)	Text chunking (MB/s)	Ratio
Snort, canonical corpus	785.4	976.1	0.80×
Snort, ~10.6 GiB	861.6	1,073.3	0.80×
Emerging Threats, canonical corpus	289.0	312.3	0.93×
Emerging Threats, ~10.6 GiB	314.5	350.0	0.90×

Source: Prepared by the author. Twelve threads in both columns; the comparator is the flat-output text chunking of Table 10.

The composition is slower than flat-output text chunking in every measured scenario, by 7% to 20%. The reason is that each shard must scan the entire input text, so K shards multiply the text traffic on the memory bus by a factor of K . In the memory-bound regime that traffic is already the bottleneck, so the extra reads outweigh any locality gained inside the smaller sub-automata. To shrink the sub-automaton below the last-level cache for the largest dictionary would require more than forty shards, multiplying the text traffic proportionally, which is unrecoverable on a memory bus of this bandwidth. This negative result is an important contribution: it closes a plausible region of the design space and reinforces the conclusion that text parallelism is the correct axis on memory-bound workloads.

6.8 END-TO-END EVALUATION

Finally, the complete pipeline, including construction, search, and merge, was measured as wall-clock time on the ~10.6 GiB corpus. Table 12 reports the best end-to-end speedup for each dictionary.

Table 12: End-to-end wall-clock speedup on the ~ 10.6 GiB corpus.

Dictionary	Sequential (ms)	Best parallel (ms)	Speedup
Snort (full)	94,706	21,158	$4.48\times$
Emerging Threats	198,906	65,428	$3.04\times$

Source: Prepared by the author. Build, search, and merge combined.

For the moderate dictionary, the full parallel stack delivers up to a $4.48\times$ end-to-end acceleration, reducing the wall-clock time from roughly ninety-five seconds to twenty-one. For the memory-bound dictionary, the best end-to-end figure is $3.04\times$, and it is obtained with flat-output text chunking and a sequential build, because under sustained load the topology-aware variant regresses sharply and the parallel build contributes less than one percent of the total time. This asymmetry is consistent with the rest of the study: in the memory-bound regime, the simplest strategy that respects the bandwidth ceiling is the most robust.

6.9 DISCUSSION AND POSITIONING

Overall, the results point to a single clear conclusion. The dominant obstacle to accelerating Aho-Corasick at intrusion-detection scale is microarchitectural: when the automaton greatly exceeds the last-level cache, performance is limited by the memory subsystem, and the most effective response is to parallelize the text while keeping the per-byte transition cost as low as possible. No single strategy is best across all regimes; the fastest strategy switches between dynamic dispatch (moderate dictionary on the canonical corpus) and the flattened variants, with and without topology awareness, in the remaining scenarios, which shows that, as the dictionary grows, the main bottleneck shifts from thread scheduling to DRAM bandwidth — the situation in which a cache-friendly emission path helps the most.

These findings can be positioned against the related literature. Compared to approaches that restructure the automaton to improve cache friendliness, such as the head-body decomposition of the matcher for multi-core intrusion detection [2], the present work attains a comparable order of magnitude of throughput while leaving the classical automaton mathematically intact, which keeps correctness easy to validate against the sequential reference. The failure-less variant of the algorithm designed for massively parallel accelerators [5] is, however, unsuitable for a multi-core CPU, because its independent per-byte traversals multiply work in a way that the memory bus cannot absorb; this is consistent with the negative result observed here for over-partitioning. Single-threaded vectorized engines that replace the automaton with SIMD literal matching [23] operate on an orthogonal axis, exploiting data parallelism within a thread rather than across threads, and are therefore complementary to, rather than competitive with, the inter-thread parallelism studied here. The finding that layout-level

changes to the emission path yield real gains aligns with work on transition-table compression for the same algorithm [21]. Finally, the use of realistic rule sets and large corpora follows the guidance on representative signature-based intrusion-detection benchmarks [6], which is what allows the memory-bound regime to be studied, rather than an artificially cache-friendly one.



7

7

CONCLUSION

This thesis investigated data-parallel scanning strategies and microarchitectural optimizations to accelerate the Aho-Corasick multi-pattern matching algorithm on shared-memory multi-core CPUs, using the POSIX Threads programming model. The work focused on a specific computational regime rather than a particular machine: the behaviour of multi-pattern matching when the automaton greatly exceeds the last-level cache, which is typical for signature-based intrusion detection with realistic rule sets.

The experiments confirmed the main hypotheses while also defining their limits. Text-level partitioning with a boundary overlap of $L_{\max} - 1$ bytes scales sublinearly, reaching $4.52\times$ at twelve threads on the multi-gigabyte corpus and a canonical-corpus peak of $4.79\times$ via dynamic dispatch. The size of the automaton, not the corpus content, proved to be the main factor for throughput: as the dictionary grows from a cache-resident set to one forty-two times larger than the last-level cache, single-threaded throughput falls by 74.0%, and the parallel speedup drops accordingly. In the memory-bound regime, the speedup peaks at $2.85\times$ near six threads (canonical corpus) and then regresses as additional threads are added: the shared DRAM bus saturates before all threads are used, making more threads counterproductive. The corresponding parallel efficiency at twelve threads is 37.7% for the moderate-dictionary regime and 23.3% for the memory-bound one, reflecting DRAM saturation as the dominant overhead rather than thread coordination. On the heterogeneous processor, frequency-weighted partitioning and dynamic dispatch removed the load imbalance of uniform segments, reducing run-to-run variation from up to 53% to under 5%. This improves predictability rather than peak throughput, since memory remains the bottleneck. The proposed flat output layout, which replaces pointer-chasing with contiguous reads, proved to be the simplest effective optimization. Its benefit grows with memory pressure, reaching +13.0% on a single thread exactly in the memory-bound regime where it matters most. Parallelizing the construction of the automaton accelerated the build of the largest dictionary by up to $1.55\times$, a useful operational gain for rule reloading, though negligible in the total time of a multi-gigabyte scan.

Equally important are the strategies that were falsified. Partitioning the dictionary instead of the text yields only a small, locality-driven gain that saturates early and never approaches the throughput of text parallelism. Combining sharding and chunking is

consistently slower than flat-output text chunking, because scanning the text once per shard multiplies the memory traffic that is already the bottleneck. These negative results are a key part of the contribution: they rule out plausible approaches and reinforce the central thesis that, on this hardware, DRAM bandwidth is the main constraint rather than the matching logic. Therefore, text parallelism combined with a cache-friendly layout is the correct approach. End-to-end, the full pipeline achieves up to $4.48\times$ on the moderate dictionary and $3.04\times$ on the memory-bound one.

The three specific objectives from the introduction were fully met. The first — to design and implement a parallel Aho-Corasick algorithm in C using the POSIX Threads API, exploiting intra-input parallelism on shared-memory multi-core CPUs — was met through a family of text-partitioning, load-balancing, and output-layout strategies whose correctness was validated exhaustively across all tested workloads. The second — to quantify performance gains in terms of speedup, throughput, and parallel efficiency against a sequential baseline — was met by reporting search speedups of $4.79\times$ (canonical corpus, dynamic dispatch) and $4.52\times$ (multi-gigabyte corpus, flat-output chunking), alongside parallel efficiencies of 37.7% and 23.3% for the moderate-dictionary and memory-bound regimes, respectively. The third — to analyze scalability under different workloads, pattern sets, and thread counts — was met through a systematic test across seven thread counts, three corpora, and automata covering a forty-two-fold range of cache capacity. This identified and quantified the memory bandwidth saturation point and the subsequent performance drop.

7.1 THREATS TO VALIDITY

The main threat to external validity is using a single mobile processor with a limited thermal envelope. This caused thermal throttling during long tests and introduced significant variance. The reported numbers are conservative under-load values, and the variability is documented instead of hidden. Crucially, the central conclusion does not depend on this machine's small cache. The largest automaton (~ 507 MiB) exceeds the last-level cache of almost all current server processors. Even on CPUs with enough total L3 cache, like AMD's V-Cache models, that capacity is split across core complexes rather than available to a single thread. Thus, the memory bandwidth bottleneck is not just an artifact of a small cache, but a property that applies to server hardware too. Testing on a smaller chip makes the issue clearer. A larger machine with uniform cores and more memory channels would delay the bottleneck, but not eliminate it. Another limitation is that the hardware-level explanations, while consistent with the timing data, were not confirmed with direct hardware telemetry.

7.2 FUTURE WORK

Two directions emerge from these limitations. The first is to use hardware perfor-

mance counters in the benchmarks. This would measure cache misses and pipeline stalls directly, confirming the memory bottleneck with hard data instead of inferring it from execution times. The second is to repeat the tests, especially the failed sharding-chunking combination, on server processors with much larger caches and higher memory bandwidth. This could reveal when dictionary sharding actually becomes beneficial. Other options include sharding based on the actual memory footprint of the sub-automata instead of pattern prefixes, and a stricter thermal protocol with cool-down periods to reduce measurement variance.

The background image shows a modern building interior with curved balconies and a central courtyard. The balconies have glass railings and are illuminated with warm lights. The courtyard features several large, cylindrical planters with green plants. The overall atmosphere is clean and contemporary.

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APPENDICES

Appendix A - Database Query Protocol

This appendix documents the protocol used to query the raw measurement database from which the search, footprint, stability, and construction results reported in this thesis are derived. The end-to-end wall-clock times of Table 12 were collected in a dedicated pipeline benchmark conducted on 2026-05-22 and are documented in the companion repository’s synthesis report (parallel-aho-corasick/docs/tcc-synthesis.html, Section 3.9); the raw sweep script and timing log are not currently archived in the repository. This allows the primary figures to be reproduced and audited independently. The benchmark campaign stores its measurements in a single typed SQLite database located at `parallel-aho-corasick/runs/overnight/sweep.db`, containing one `runs` table (one row per timed execution) and a set of aggregating views. The headline figures are retrieved from these views instead of the raw execution logs.

9.1 DATABASE SCHEMA

Each row of the `runs` table identifies an execution by its phase, pattern dictionary, corpus, searcher strategy, and thread count. It records the timing statistics (minimum, mean, median, and maximum wall-clock time in milliseconds, along with the coefficient of variation across the timed iterations), the derived throughput (in MB/s and Gbit/s), the input volume and match count, and the automaton descriptors (pattern count, number of states, footprint in kibibytes, and sequential or parallel build time). The views listed in Table 13 pre-aggregate the data so no headline figure requires ad hoc post-processing of the raw rows.

Table 13: Aggregating views of the measurement database.

View	Question it answers
<code>v_speedup</code>	Speedup of each searcher against the single-threaded sequential baseline.
<code>v_self_speedup</code>	Self-relative scalability: throughput at T threads over throughput at one thread.
<code>v_footprint</code>	Throughput as a function of the automaton footprint, at one and twelve threads.
<code>v_build</code>	Speedup of the parallel automaton construction over the sequential build.
<code>v_best</code>	Best-performing searcher for each dictionary, corpus and thread count.
<code>v_correctness</code>	Number of distinct match counts per dictionary and corpus (a value of one indicates agreement).

Source: Prepared by the author.

9.2 CANONICAL QUERIES

The queries below reproduce the main results of the study. They can be executed against the database using the standard `sqlite3` client.

```
-- Speedup curve of a searcher (Snort dictionary, canonical corpus)
```

```
SELECT thr, speedup_vs_seq FROM v_speedup
WHERE patterns='patterns_snort' AND corpus='enron_corpus'
AND searcher='pthread_dynamic' ORDER BY thr;
```

```
-- Peak speedup per scenario (winning searcher and thread count)
```

```
SELECT patterns, corpus, searcher, thr, speedup_vs_seq FROM v_speedup w
WHERE speedup_vs_seq = (SELECT MAX(speedup_vs_seq) FROM v_speedup x
WHERE x.patterns = w.patterns AND x.corpus = w.corpus)
ORDER BY patterns, corpus;
```

```
-- Footprint versus throughput at twelve threads (cache blowout)
```

```
SELECT patterns, automaton_mib, searcher, mbps FROM v_footprint
WHERE thr = 12 ORDER BY automaton_mib;
```

```
-- Speedup of the parallel automaton construction
```

```
SELECT patterns, build_threads, build_speedup FROM v_build
ORDER BY patterns, build_threads;
```

Before any timing was reported, the correctness of every parallel strategy was verified using the `v_correctness` view. This view returns a single distinct match count for each dictionary and corpus, confirming that all strategies reproduce the exact match set of the sequential baseline.



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